

The Fertility, Marriage, and Gender Equality Quandary

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Abstract

This paper documents that societies achieving any two of high fertility, dual parenthood, and gender income equality almost invariably fall short on the third. Data from 115 countries reveal that this shortfall is equally large in OECD and non-OECD countries but arises from different tensions: in the OECD, high fertility and dual parenthood seldom coincide; elsewhere, the central conflict is between fertility and gender equality. A marriage-market model in which marriage bundles income sharing with joint fertility rationalizes both patterns. While conventional policies each improve two objectives at the expense of the third, coordinating multiple instruments can overcome this trade-off but requires delicate balancing. A simpler alternative — redistributing childcare from wives to husbands — succeeds when both spouses gain in the equilibrium. A calibration to Mexico quantifies the mechanism.

JEL classification: D13, J11, J12, J13, J16

Keywords: Gender equality, family structure, fertility

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1 Introduction

Three demographic transformations have reshaped advanced economies over the past half-century. Fertility has fallen, often well below replacement. Stable two-parent families have grown less common, and a rising share of children now spend part of their childhood in single-parent households. Meanwhile, gender gaps in education and earnings have narrowed substantially, and in some settings reversed (Goldin, 2014).

Each transformation has motivated a distinct policy agenda. Concerns about pension sustainability (Bongaarts, 2004) and long-run economic growth (Jones, 2022) have prompted pro-fertility measures ranging from baby bonuses to subsidized childcare. A long-standing interest in family stability, rooted partly in cultural considerations and partly in evidence on the consequences of single parenthood for children’s outcomes (Autor et al., 2019; Wasserman, 2020), has motivated pro-marriage policies such as joint taxation. Efforts to promote gender equality, meanwhile, seek both to advance equal opportunity and to correct the misallocation of human capital, whose aggregate costs can be substantial (Hsieh et al., 2019). These agendas are usually discussed separately. Yet the decisions they seek to influence are closely connected: having children changes parents’ market work, earning opportunities shape the attraction of marriage, and marriage changes the resources available for raising children. When does progress toward one objective support the others, and when does it come at their expense? I address this question in four steps.

First, I construct a global panel that combines the best available measures for each dimension: the total fertility rate from the World Development Indicators, children’s co-residence with both parents from census microdata and the OECD Family Database, and the female share of labor income from the World Inequality Database. The resulting sample covers 115 countries and 1,764 country-year observations. Societies that succeed on any two of high fertility, dual parenthood, and gender income equality almost invariably fall short on the third. Only 2.8 percent of observations achieve all three, far below the 12.5 percent implied by statistical independence. To understand the source of this shortfall, I divide the sample into high-income (proxied by OECD) and low-income (non-OECD) countries. While the magnitude of the shortfall is remarkably similar across the two subsamples (about 6 percentage points in each), its composition differs sharply. In OECD countries, the dominant tension is between fertility and dual parenthood: high fertility and widespread two-parent families rarely coincide. In non-OECD countries, the dominant tension is between fertility and gender income equality. These contrasting patterns, which an exact pairwise decomposition makes precise, call for a unified framework that can accommodate the distinct bottlenecks facing different groups of countries.

Second, I develop a tractable equilibrium marriage-market model that connects the three outcomes through income sharing and the time cost of children. Men and women draw idiosyncratic tastes for marriage. Marriage bundles an income transfer from husband to wife with shared fertil-

ity: the wife chooses fertility conditional on the transfer, and the husband evaluates the resulting consumption–fertility bundle when deciding whether to marry. The transfer then adjusts to balance participation on the two sides of the market. Fertility and the transfer are therefore jointly determined.

The model weaves together three outcomes of interest: aggregate fertility, marriage, and the gender income gap. When mothers provide more childcare, higher fertility reduces relative female labor supply. At the same time, the income-sharing value of marriage depends on women’s earning capacity relative to the support available from husbands. Their interaction generates a simple yet important policy tension. A larger marriage benefit or a lower cost of children raises fertility and marriage but widens the gender income gap. Higher female wages narrow the gap but reduce fertility and marriage. The link between fertility and gender income inequality through maternal time use is well documented ([Kleven et al., 2019](#)). The marriage market introduces an additional channel, because the equilibrium transfer responds endogenously to policy. A reform’s effect on any one outcome therefore depends on how the marital contract adjusts on the other two.

Importantly, the model also offers an elegant explanation as to why the dominant tension differs across country groups. The gender wage gap is the key state variable. When the gap is large, as in most non-OECD economies, each unit of fertility strongly amplifies gender income inequality through the time cost of children, making the fertility–gender equality tension dominant. Simultaneously, the large marital fertility premium ties marriage and fertility tightly together through the income-sharing channel, preventing them from conflicting. When the gap is smaller, as in the OECD, that amplification weakens: fertility and gender equality decouple, the composition channel linking marriage to fertility loosens, and the tension between fertility and dual parenthood emerges as the principal bottleneck. These predictions match the decomposition and connect the model’s core mechanism to the strikingly different sources of the shortfall in high- and low-income countries.

Third, I ask whether policy can loosen this three-way trade-off. I show that three standard levers (subsidizing marriage, closing the wage gap, and lowering the overall cost of children) each improve two of the three outcomes at the expense of the third. I then ask whether combining instruments can overcome this limitation. A carefully chosen mix of higher female wages and a larger marriage benefit can, in principle, raise all three outcomes simultaneously. The marriage benefit can be calibrated to offset the fertility decline that higher female wages would otherwise cause, while preserving their income-gap benefits. The result establishes that the quandary is not inherent to the economic environment but rather a feature of policy instruments considered in isolation. However, such coordination requires delicate balancing: the marriage-benefit adjustment must be finely tuned to the wage change, and any imbalance leaves at least one objective worse off.

This practical difficulty motivates the search for a simpler, single-instrument approach. I con-

sider a *redistribution* of childcare obligations from wives to husbands, an increasingly popular policy lever used in developed economies to promote gender equality in childcare (Farré and González, 2019; Patnaik, 2019). The baseline asymmetry that such policies target is well documented: mothers shoulder far more of the time burden than fathers. Viewed through a wider lens, however, this asymmetry is itself plausibly a legacy of traditional gender norms rather than what technology or preferences would have justified (Goldin, 2025; Deschamps et al., 2026; Albrecht et al., 2024; Xi and Zhou, 2026). Redistribution, such as through mandated paternity leaves or changes in social norms, can therefore be understood as a corrective that pushes toward a more efficient childcare allocation.

I show that childcare redistribution can raise fertility, marriage, and gender income equality simultaneously when three conditions hold. The first requires redistribution to lower the wife’s effective childcare cost, raising marital fertility. The second requires the shift in childcare to improve women’s relative labor supply even after the additional fertility it induces creates more care to be provided. The third concerns the male side of the market: despite providing more childcare, the husband must gain after fertility and transfers adjust in equilibrium. These conditions become more demanding when fathers are less efficient than mothers at providing care. The model captures this possibility by allowing each unit of the father’s time to produce only $\eta \in (0, 1]$ effective units of childcare. Lower efficiency raises the father’s time cost per unit of care and tightens all three conditions. A minimum efficiency threshold $\bar{\eta}$, below which at least one condition fails, provides a transparent summary of how much room for redistribution exists in a given economic environment.

The requirement that both spouses gain connects to a prominent conjecture in demography. Goldscheider et al. (2015) argue that the second stage of gender equality, in which men take on greater domestic roles, can restabilize families. The analysis here identifies when this conjecture holds and when it does not. Restabilization requires not only that the wife’s effective childcare cost falls but also that the resulting marital contract remains attractive to the husband after fertility and transfers adjust. This bilateral requirement is the paper’s central theoretical point: mothers’ time constraints and fathers’ willingness to enter the marital contract jointly shape the equilibrium response to policy. Relieving women’s constraints raises marriage and fertility only when the resulting contract continues to offer men a sufficiently attractive exchange between consumption and children. Bringing both sides into the analysis explains why similar policies can produce different demographic responses across settings.

Fourth, I bring the framework to the data by calibrating the extended model, in which both spouses provide childcare, to Mexico from 1990 to 2025.¹ Three time-varying inputs drive the calibration: gender-neutral productivity, the relative gender wage gap, and the evolving division of

¹Mexico offers a long time series, substantial wage changes during its demographic transition, and a trajectory broadly representative of transitions in many middle-income economies.

childcare between mothers and fathers, the last constructed from Mexico’s National Survey on Time Use (ENUT). The period features falling fertility, rising single motherhood, and narrowing gender earnings gaps. With four estimated parameters, the model closely tracks fertility and the child-based measure of family structure. As an out-of-sample check, the model’s predicted female labor supply closely tracks Mexico’s female labor force participation rate from FRED, reproducing both the rapid increase through 2008 and the subsequent deceleration. A Shapley-value decomposition, which assigns each force its average marginal contribution across all possible orderings, attributes the bulk of the fertility, family-structure, and welfare changes to the declining relative wage gap, while the increasingly equal division of childcare partially offsets the fertility and dual-parenthood declines. Consumption-equivalent welfare measures show that all agent types converge toward married-male welfare over the sample period, driven almost entirely by the closing wage gap. Applying the calibrated model to the paper’s central policy experiment, I find that all three conditions for childcare redistribution to improve the trinity are satisfied throughout 1990–2025. The minimum paternal efficiency threshold declines from approximately 0.94 in 1990 to roughly 0.77 by the mid-2010s, indicating that the scope for redistribution widens as wages converge.

Appendix F explores a dynamic extension in which lower marriage reduces the next generation’s male relative human capital, motivated by evidence that boys are more sensitive than girls to the absence of a biological father (Sommers, 2001; Autor et al., 2019; Wasserman, 2020). Whether the resulting feedback amplifies the demographic response or fades across generations remains a quantitative question.

Related Literature

The paper connects several literatures. A long tradition links fertility decisions to female labor supply through the time cost of children, beginning with Rosenzweig and Wolpin (1980) and continuing in quantitative models such as Erosa et al. (2016) and Blundell et al. (2016). Feyrer et al. (2008) document the reversal of the cross-country correlation between female labor force participation and fertility, suggesting that institutional adaptation can reconcile the two. Goldscheider et al. (2015) propose a gender revolution framework in which progress toward gender equality first destabilizes and then restabilizes families as men take on greater domestic roles, a mechanism that the redistribution analysis in this paper formalizes. Related work emphasizes household production, childcare marketization, and parental leave (Bar et al., 2018; Greenwood, 2019; Kim and Yum, 2025); Doepke et al. (2023) and Greenwood et al. (2023) provide surveys. Doepke and Kindermann (2019) show that spousal disagreement over fertility is quantitatively important under limited commitment, emphasizing the bilateral nature of reproductive decisions. I build on these mechanisms by placing them inside a marriage market, where the marital contract adjusts to policy, linking changes in women’s constraints to men’s willingness to marry.

A second literature, originating with [Becker \(1973\)](#), studies marriage-market clearing when men and women differ in productivity and preferences. Transferable-utility matching frameworks ([Choo and Siow, 2006](#); [Chiappori, 2017](#)) have been used to examine divorce law, household production, and technological change ([Stevenson and Wolfers, 2007](#); [Voena, 2015](#); [Greenwood et al., 2016](#); [Low, 2024](#)). I abstract from within-gender wage heterogeneity to isolate the between-gender margin and use an imperfectly transferable utility framework with micro-founded spousal utilities, following [Bisin and Tura \(2019\)](#). Changing the transfer changes fertility as well as consumption, making the marital bundle and the spouses' resulting utilities endogenous to the terms of income sharing.

A third literature studies how technological change reshapes the family ([Greenwood et al., 2005a, 2016](#); [Ngai and Petrongolo, 2017](#); [Cao et al., 2024](#)). [Olivetti and Petrongolo \(2016\)](#) and [Feng et al. \(2023\)](#) document long-run gender-gap changes, while [Hsieh et al. \(2019\)](#) quantify gains from improved talent allocation. I decompose wage changes into gender-neutral and gender-biased components and connect them to fertility, marriage, and income inequality through a common equilibrium.

A fourth literature studies how family structure shapes children's human capital, with boys appearing particularly vulnerable to the absence of a biological parent ([Sommers, 2001](#); [Bertrand and Pan, 2013](#); [Chetty et al., 2016](#); [Autor et al., 2019](#); [Wasserman, 2020](#); [Reeves, 2022](#); [Autor et al., 2023](#); [Frimmel et al., 2024](#)). [Kearney \(2023\)](#) discusses the implications for inequality and mobility. Appendix [F](#) uses this evidence to motivate a feedback from current marriage to future male relative human capital.

Empirically, the paper uses a descriptive benchmark in the spirit of the dartboard logic of [Ellison and Glaeser \(1997\)](#): joint attainment is compared with the frequency implied by independent outcomes having the same observed marginal attainment rates. To move beyond the aggregate shortfall, I decompose it into contributions from each pairwise tension using an exact algebraic identity, identifying which specific bilateral conflicts are the dominant bottlenecks in different groups of countries. These tools organize the empirical motivation for studying the outcomes together; the model supplies the economic structure needed to examine their responses to policy.

The rest of the paper is organized as follows. Section [2](#) presents the empirical patterns. Section [3](#) develops the baseline model, individual-policy comparative statics, and the coordinated-policy result. Section [4](#) studies childcare redistribution. Section [5](#) presents the quantitative analysis. Section [6](#) concludes. Appendix [F](#) develops a dynamic extension with cross-generational feedback.

2 Motivating Facts

This section establishes the three-way tension that motivates the theory. I assemble a global panel using the best available measures for each dimension, document that the trinity of high fertility,

dual parenthood, and gender income equality is rarely achieved, and decompose the shortfall to reveal its contrasting sources in high- and low-income countries.

2.1 Data Sources and Construction

Data sources

Three categories of data, drawn from four source datasets, are combined to construct the analysis sample:

- (i) **Total fertility rate** from the World Development Indicators (WDI), measured as children per woman.
- (ii) **Dual parenthood** from two complementary sources. The IPUMS International (IPUMSI) database provides census microdata on children’s household composition, from which I compute the share of children living with both parents. The OECD Family Database provides analogous national-level statistics on children’s living arrangements. For country-years with data from both sources, I average the two measures; otherwise I use whichever is available.
- (iii) **Gender income equality**, measured as the female share of national labor income from the World Inequality Database (WID) ([Chancel et al., 2022](#)). This measure is more comprehensive than the earnings gap among workers, since it incorporates differences in labor-force participation.²

Panel construction

I build a master country-year panel by taking the union of all country-years appearing in any of the four source datasets, then merge the variables by country and year. Missing values between observed data points within a country are filled by linear interpolation. The analysis retains only complete cases: country-years for which all three variables (fertility, dual parenthood, and female income share) are observed or interpolated.

Sample overview

The resulting *full sample* contains 1,764 country-year observations across 115 countries. Data availability for the dual-parenthood measure naturally divides the sample into two groups. Countries for which at least one year of OECD Family Database data exists form the *OECD subsample* (879 observations, 39 countries). The remaining countries, whose dual-parenthood data come

²Replacing the WID female income share with the OECD unadjusted gender wage gap among employed workers reaches similar conclusions (Appendix [A.3](#)).

exclusively from IPUMSI census microdata, form the *non-OECD subsample* (885 observations, 76 countries). Table 1 summarizes the three samples.

Sample	Countries	Obs	Dual parenthood source	Gender equality measure
Full	115	1,764	IPUMSI + OECD	Female income share (WID)
OECD	39	879	OECD Family Database (+ IPUMSI)	Female income share (WID)
Non-OECD	76	885	IPUMSI only	Female income share (WID)

Table 1. Summary of Analysis Samples

Notes: All samples use the total fertility rate from the WDI for fertility and the female labor income share from the WID for gender equality. The OECD subsample includes countries with at least one year of OECD Family Database dual-parenthood data. Where both IPUMSI and OECD data are available for a country-year, the two are averaged. Missing values are interpolated linearly within country between observed data points.

The choice of measures is deliberate. The total fertility rate is the standard demographic indicator of fertility. Children’s co-residence with both parents, constructed from household-level census data, directly captures the family environment relevant for child development and avoids the complications of using marriage or nonmarital-birth statistics, which can differ substantially from co-residence patterns. The female income share accounts for both wages and participation, providing a comprehensive measure of gender economic equality.

2.2 The Trinity Shortfall

Variable construction

For each sample, I split all three outcomes at their sample-specific medians. “High fertility” (F) denotes a total fertility rate exceeding the sample median. “Dual parenthood” (M) denotes a share of children with both parents exceeding the sample median. “Gender equality” (G) denotes a female income share exceeding the sample median. Using sample medians avoids arbitrary thresholds and yields approximately balanced categories. All classifications use strict inequalities: an observation exactly at its cutoff remains outside the corresponding attainment group.

Table 2 reports the cutoffs for all three samples. Their variation highlights the heterogeneity that motivates the subsample analysis. The median TFR ranges from 1.56 in the OECD to 3.24 in non-OECD countries. The dual-parenthood median ranges from 70.5 percent in non-OECD countries to 83.7 percent in the OECD. The female income share median ranges from 29.4 percent in non-OECD to 37.4 percent in OECD countries.

The benchmark asks how often all three indicators would equal one if their observed marginal attainment rates were combined independently. Writing p_F , p_M , and p_G for those rates, expected joint attainment is $p_{111}^{\text{ind}} = p_F p_M p_G$. Independent indicators with marginal attainment probabilities of one-half yield 12.5 percent.

Sample	High fertility (F)	Dual parenthood (M)	Gender equality (G)
Full	TFR > 2.00	Both parents > 80.1%	Female share > 34.1%
OECD	TFR > 1.56	Both parents > 83.7%	Female share > 37.4%
Non-OECD	TFR > 3.24	Both parents > 70.5%	Female share > 29.4%

Table 2. Median Cutoffs for Binary Indicators

Notes: Displayed cutoffs are rounded; calculations use the numerical thresholds returned by the statistical routine. TFR denotes the total fertility rate, children per woman. “Both parents” is the share of children living with both parents. “Female share” is the female share of national labor income. Appendix Table A.1 reports the exact cutoffs alongside additional sample statistics.

Full-sample results

Figure 1 presents the Venn diagram for the full sample. Only 2.8 percent of country-year observations achieve the trinity F+M+G, compared with the 12.5 percent independence benchmark. The two largest categories are “F only” (26.9 percent) and “M+G” (25.7 percent), revealing a stark divide: countries with high fertility tend to lack dual parenthood and gender equality, while countries with dual parenthood and gender equality tend to have low fertility.

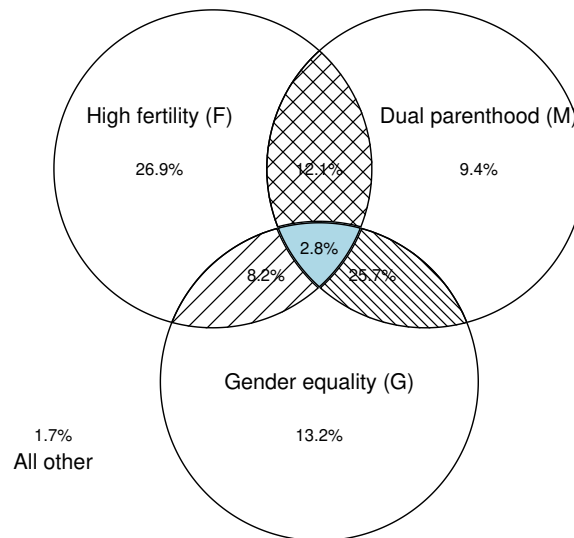


Figure 1. Three-Way Trade-Off in the Full Sample ($N = 1,764$)

Notes: Venn diagram showing the joint distribution of high fertility (F), dual parenthood (M), and gender equality (G), each defined at the full-sample median. Under statistical independence, the central overlap (F+M+G) would contain 12.5 percent of observations. The empirical independence benchmark is 12.49 percent.

Country-clustered bootstrap inference (1,000 replications) yields a 95 percent percentile interval of $[-11.87, -6.48]$ percentage points for the shortfall, with a normal-approximation p -value below

0.001. The trinity is significantly rarer than independence would predict.

Motivation for subsample analysis

The full sample pools countries with total fertility rates ranging from below 1 to nearly 7, dual-parenthood shares from 26 to 96 percent, and female income shares from 3 to 50 percent. Median splits within such a heterogeneous pool may conflate differences in levels with genuine trade-offs within comparable groups of countries. To address this concern, I split the sample into OECD and non-OECD countries, re-estimating all medians and binary indicators within each subsample.

2.3 OECD and Non-OECD Results

Figure 2 presents Venn diagrams for the two subsamples. Table 3 reports the trinity shortfall for all three samples.

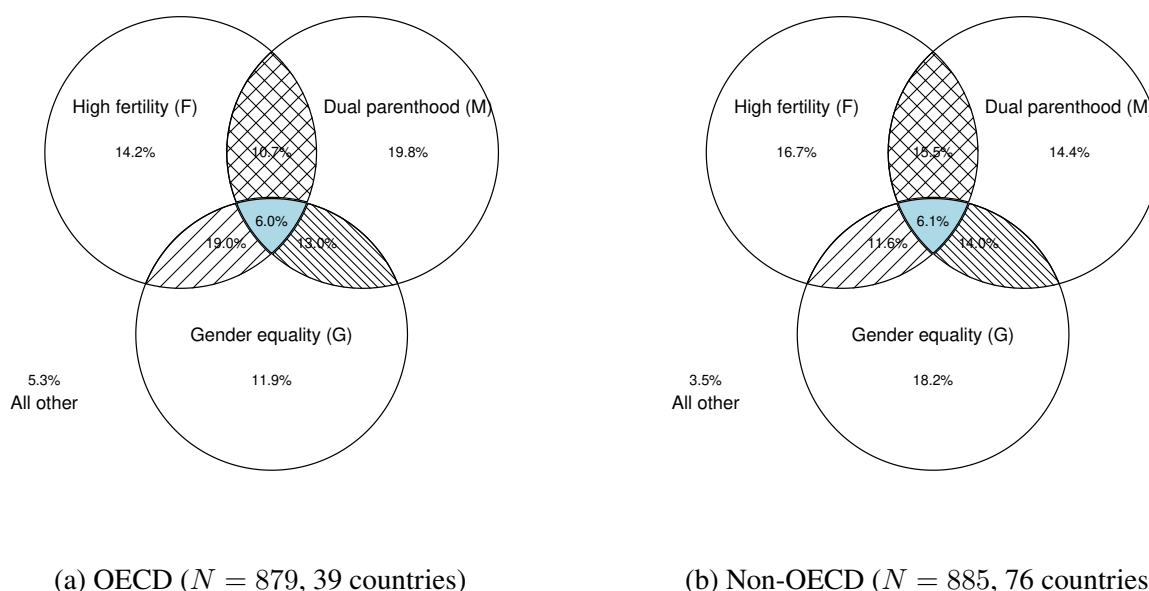


Figure 2. Three-Way Trade-Off: OECD and Non-OECD Subsamples

Notes: Each subsample uses its own median cutoffs. Under statistical independence, the central overlap would contain approximately 12.5 percent. See Table 3 for bootstrap inference.

Two findings stand out. First, the shortfall is present and statistically significant in both subsamples, despite the very different country compositions and cutoff levels. Second, the magnitude of the shortfall is remarkably similar: 6.34 percentage points in the OECD and 6.38 percentage points in non-OECD countries, with observed-to-independence ratios of 0.49 in both cases.³

³The full-sample shortfall (9.65 percentage points) is larger than either subsample's, reflecting additional compositional dependence created by pooling countries at very different levels of development.

Sample	Observed (%)	Independence (%)	Shortfall (pp)	Ratio	95% CI (pp)	<i>p</i> -value
Full	2.83	12.49	−9.65	0.23	[−11.87, −6.48]	< 0.001
OECD	6.03	12.37	−6.34	0.49	[−10.34, −2.19]	0.002
Non-OECD	6.10	12.49	−6.38	0.49	[−9.85, −0.98]	0.005

Table 3. Trinity Shortfall Across Samples

Notes: Shortfall is observed minus independence benchmark joint-attainment share. Ratio is observed divided by independence benchmark. Bootstrap inference uses 1,000 replications clustered by country. Percentile confidence intervals are reported. The *p*-value is the two-sided normal approximation based on the observed coefficient and bootstrap standard error.

While the levels of the shortfall coincide, the Venn diagrams hint at different underlying patterns. In the OECD, the largest single category is “M only” (19.8 percent), representing countries that achieve dual parenthood but fail on fertility and gender equality, alongside sizable “F+G” (19.0 percent) and “F only” (14.2 percent) categories. In non-OECD countries, the distribution is more dispersed, with “G only” (18.2 percent), “F only” (16.7 percent), and “F+M” (15.5 percent) all prominent. These differences point to contrasting pairwise tensions that the next subsection quantifies.

2.4 Decomposing the Shortfall

A brief primer on the method

To move beyond the aggregate shortfall, I decompose it into contributions from each pairwise association. The tool is an exact algebraic identity. For any three binary variables, the deviation of joint attainment from the independence benchmark can be written as

$$p_{111} - p_F p_M p_G = \underbrace{\delta_{FM} \cdot p_G}_{\text{F-M contribution}} + \underbrace{\delta_{FG} \cdot p_M}_{\text{F-G contribution}} + \underbrace{\delta_{MG} \cdot p_F}_{\text{M-G contribution}} + \varepsilon, \quad (1)$$

where $\delta_{XY} = P(X=1, Y=1) - P(X=1)P(Y=1)$ measures the pairwise excess for each pair, and ε is a residual capturing genuine three-way interaction. A negative δ_{XY} means the two outcomes X and Y are negatively associated (they “repel” each other), and the corresponding contribution is negative, pulling the trinity share below what independence would predict. Each pairwise contribution is weighted by the marginal probability of the third dimension.

This decomposition is related to the broader framework of mutual information, an information-theoretic measure of statistical dependence. Mutual information quantifies how much knowing one variable reduces uncertainty about another, capturing dependencies of any form without distributional assumptions. The decomposition in (1) provides a direct, interpretable link between pairwise associations and the trinity shortfall, while the mutual information framework provides

complementary diagnostics. Appendix A.2 reports the full set of bootstrap results, including mutual information statistics.

Different tensions in different settings

Table 4 presents the decomposition for the OECD and non-OECD subsamples.

Component	OECD		Non-OECD	
	Contribution (pp)	% of shortfall	Contribution (pp)	% of shortfall
F-M tension	−4.02	63.4%	−1.71	26.7%
F-G tension	+0.07	−1.1%	−3.57	55.9%
M-G tension	−2.86	45.1%	−2.42	37.9%
Three-way residual	+0.47	−7.4%	+1.31	−20.5%
Total shortfall	−6.34	100%	−6.38	100%

Table 4. Pairwise Contribution Decomposition of the Trinity Shortfall

Notes: The identity in (1) decomposes the total shortfall into contributions from each pairwise tension plus a three-way residual. A negative contribution means the corresponding pair of outcomes repels each other, making the trinity rarer. A positive contribution means the pair tends to co-occur, partially offsetting the shortfall. Bootstrap inference is reported in Appendix Table A.4.

The contrast is striking. In OECD countries, the dominant tension is between fertility and dual parenthood (F-M), accounting for 63.4 percent of the shortfall: high fertility and widespread two-parent families rarely coincide. The tension between dual parenthood and gender equality (M-G) accounts for a further 45.1 percent. Together, these two tensions place dual parenthood as the bottleneck dimension, since it is in conflict with both of the other objectives. Remarkably, the fertility–gender equality pair (F-G) contributes essentially nothing (−1.1 percent): in the OECD, fertility and gender income equality are approximately uncorrelated.

In non-OECD countries, the picture reverses. The dominant tension is between fertility and gender equality (F-G), accounting for 55.9 percent of the shortfall. The M-G tension contributes 37.9 percent, and the F-M tension only 26.7 percent. Gender equality is the bottleneck dimension: it conflicts with both fertility and dual parenthood. All three pairwise tensions contribute negatively, and the three-way residual is positive, partially offsetting the pairwise effects.

Figure 3 visualizes this cross-sample comparison. Bootstrap confidence intervals (Appendix Table A.4) confirm that the F-M tension is significantly negative in the OECD ($p = 0.003$) and the F-G tension is significantly negative in the non-OECD ($p = 0.005$), while the F-G tension in the OECD is indistinguishable from zero ($p = 0.964$).

These OECD patterns are robust to an alternative gender equality measure that removes the mechanical link between fertility and measured income inequality through labor-force participation. Replacing the WID female income share with the OECD unadjusted gender wage gap among

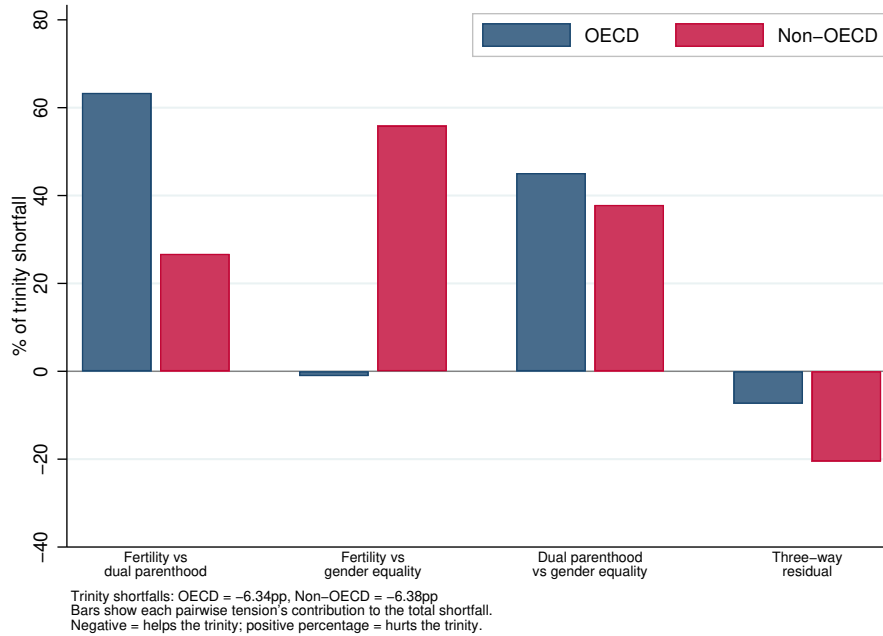


Figure 3. Decomposition Comparison: OECD vs Non-OECD

Notes: Bars show each pairwise tension's contribution to the total trinity shortfall, expressed as a percentage of the shortfall. A positive bar means the pair's negative association accounts for a share of the shortfall; a negative bar means the pair's positive association partially offsets it. Bootstrap inference is reported in Appendix Table A.4.

employed workers, the fertility–dual parenthood tension accounts for virtually the entire shortfall, while the fertility–gender equality tension remains negligible (Appendix A.3).

2.5 Summary

The empirical analysis yields two findings. First, simultaneously achieving high fertility, widespread dual parenthood, and gender income equality is rare across 115 countries spanning the global income distribution. The joint share is far below the independence benchmark, and the shortfall is statistically significant in both OECD and non-OECD subsamples. Second, while the magnitude of this shortfall is nearly identical in high- and low-income countries, its composition differs sharply: the fertility–dual parenthood tension dominates in the OECD, while the fertility–gender equality tension dominates in non-OECD countries.

These findings speak to contemporary debates about the transformation of the family. Across the world, fertility is declining, dual parenthood is becoming less common, and gender income gaps are converging. Commentators increasingly discuss the “decoupling” of fertility from marriage alongside the potential compatibility of fertility and gender equality. The decomposition confirms both observations as features of the OECD data: fertility and gender equality are essentially unrelated there, while fertility and dual parenthood are strongly in tension. In developing countries,

however, the classic conflict between fertility and women’s economic participation remains the primary constraint.

The decomposition documents which pairwise tensions are most severe but cannot explain why they differ across country groups or whether policy can loosen them. Answering both questions requires a framework in which fertility, marriage, and the gender income gap are jointly determined—so that a change in one feeds back through the others—and in which the terms of the marital contract adjust endogenously to policy. The next section provides such a framework.

3 Model

This section develops an equilibrium marriage-market model in which fertility, marriage, and the gender income gap are determined together. The model is deliberately simple. Its purpose is to make clear how the time cost of children and the income-sharing role of marriage connect the three outcomes, and why policies directed at one can move the others in an unintended direction.

3.1 Setup

Individuals are indexed by gender $g \in \{\sigma, \varphi\}$, with each gender comprising half the population. Utility depends on consumption c and fertility n through a constant-elasticity-of-substitution aggregator:

$$u(c, n) = \left[(1 - \beta)c^{\frac{\rho-1}{\rho}} + \beta n^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad \beta \in (0, 1), \quad \rho > 1. \quad (2)$$

Following [Jones and Schoonbroodt \(2010\)](#) and [Carlos Córdoba and Ripoll \(2019\)](#), I assume that consumption and fertility are substitutes. This assumption allows rising opportunity costs to reduce fertility as wage grows. Utility in (2) is also well-defined for childless individuals, with $u(c, 0) = (1 - \beta)^{\rho/(\rho-1)}c$.

To focus on between-gender differences, wages are uniform within each gender and strictly positive.⁴ Let w^σ and w^φ denote male and female wages, and define the gender wage gap as $\Gamma^w = w^\sigma / w^\varphi$. A higher ratio indicates a larger male wage advantage. Wages are exogenous in the static model.⁵

⁴Abstracting from within-gender wage and education heterogeneity shuts down assortative matching, which would require additional structure ([Chiappori, 2017](#)). Figure A.3 reports a spousal earnings gap among newly-wed, childless U.S. couples of 20 to 30 percent, compared with approximately 10 percent among young workers, motivating attention to within-couple gender differences.

⁵In life-cycle models such as [Erosa et al. \(2016\)](#) and [Blundell et al. \(2016\)](#), labor supply also affects subsequent human capital and wages. The static model isolates the contemporaneous labor-supply and marriage-market channels.

Single individuals

Single men supply one unit of labor, consume their earnings, and remain childless:

$$V^{\sigma^{\varnothing},s} = u(w^{\sigma^{\varnothing}}, 0).$$

Single women can have children without support from the father.⁶ They choose consumption, market work, and fertility to solve

$$V^{\varnothing,s} = \max_{c^{\varnothing,s}, l^s, n^s} u(c^{\varnothing,s}, n^s)$$

subject to

$$c^{\varnothing,s} = w^{\varnothing} l^s, \quad l^s = 1 - \chi n^s, \quad c^{\varnothing,s} \geq 0, \quad l^s \geq 0, \quad n^s \geq 0.$$

Each child requires $\chi > 0$ units of the mother's time. A single mother therefore faces a direct choice between market consumption and children: more children leave less time to earn the income that finances consumption. Following [De La Croix and Doepke \(2003\)](#), fertility is continuous.

The setup captures the competition between consumption and fertility for household resources. Purchased childcare ([Bar et al., 2018](#)) and explicit investments in child quality ([Becker and Lewis, 1973](#)) are left for extensions. Section 4.5 discusses how outsourced care would differ from the time-cost changes studied here.

Married individuals

Marriage changes the resources available for raising children. In the baseline, husbands provide no childcare, supply one unit of labor, and transfer a share α of their earnings to their wives. The transfer captures the income-sharing function of marriage. Individuals take it as given, while its equilibrium value equates the number of men and women willing to marry.

The necessary resource restrictions are

$$-\frac{w^{\varnothing}}{w^{\sigma^{\varnothing}}} < \alpha < 1.$$

The lower bound ensures positive resources for the wife before childcare, and the upper bound leaves positive consumption for the husband. Negative transfers are permitted in principle, although the

⁶The model abstracts from child support. Introducing enforceable support obligations would change both the resources available to single mothers and men's outside option, and would therefore require a corresponding change in the marriage-market analysis. According to the analysis by the Annie E. Casey Foundation using 2020-22 Current Population Survey data, only 23 percent of U.S. female-headed families with children under 18 receive any child support in the prior year.

interior equilibrium characterized below has $\alpha > 0$.

Fertility decision rights are assigned to the wife. She chooses fertility conditional on the committed transfer, and the husband evaluates the resulting bundle when deciding whether to marry.⁷ His marital value is

$$V^{\sigma^*,m} = \lambda u((1 - \alpha)w^{\sigma^*}, n^m), \quad \lambda > 0.$$

The wife solves

$$V^{\varnothing,m} = \max_{c^{\varnothing,m}, l^m, n^m} \lambda u(c^{\varnothing,m}, n^m)$$

subject to

$$c^{\varnothing,m} = \alpha w^{\sigma^*} + w^{\varnothing} l^m, \quad l^m = 1 - \chi n^m, \quad c^{\varnothing,m} \geq 0, \quad l^m \geq 0, \quad n^m \geq 0.$$

Parameter λ is a utility shifter that scales the utility from marriage, reflecting factors such as social norms. In the baseline model, I assume that λ is gender-neutral.⁸

The within-marriage transfer creates an income effect for the wife: at a given wage, more support allows her to choose both more consumption and more children. A higher female wage raises income while also making time spent caring for children more expensive. With $\rho > 1$, the opportunity-cost effect dominates and desired fertility falls at a fixed nonnegative transfer. This is the substitution mechanism emphasized by [Greenwood et al. \(2005a\)](#). A higher male wage raises the resources available for child-rearing at a fixed positive transfer, consistent with the income channel in [Jakobsen et al. \(2022\)](#). The equilibrium analysis adds a further step by allowing the transfer itself to respond to these wage changes.

Marriage market

At the start of the period, each individual draws independent, multiplicative taste shocks for marriage and singlehood. The shocks have a common Fréchet distribution with shape parameter $\theta > 0$. Individuals choose the alternative with the higher taste-adjusted value, yielding

$$\mathcal{M}^g = \frac{(V^{g,m})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta}, \quad g \in \{\sigma^*, \varnothing\}. \quad (3)$$

⁷The timing commits the transfer before fertility is chosen and assigns the fertility decision to the wife. Introducing subsequent bargaining, consent, or renegotiation after children are born would require a different contracting environment.

⁸Allowing for gender-specific λ would not change the analysis as the main propositions come from comparative statics. With additional data, such as direct measures of within-marriage transfers, it would be feasible to calibrate gender-specific λ .

A higher marriage-to-singlehood utility ratio attracts more participants; a larger θ makes participation more responsive to that ratio. Appendix D.1 derives (3) directly from the taste-shock distribution.

With equal masses of men and women, market clearing requires $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$. Because the two genders have the same taste-shock distribution, this is equivalent to

$$\frac{V^{\sigma,m}}{V^{\sigma,s}} = \frac{V^{\varphi,m}}{V^{\varphi,s}}. \quad (4)$$

Equation (4) is the key equilibrium condition. The terms of marriage must make it equally attractive relative to singlehood on the two sides of the market. The transfer adjusts to achieve this balance, taking account of the fertility response it induces. Together, (α, n^m) describes the marital bundle: the transfer clears the market, and the wife's household choice determines fertility conditional on that transfer.

The competitive interpretation follows Choo and Siow (2006): individuals are atomistic and take the terms of marriage as given. Here, changing α changes the bundle itself. A larger transfer changes both the division of household resources and the number of children, so both spouses' valuations respond to the wife's fertility choice.

Aggregate outcomes

The three outcomes of interest are aggregate fertility, marriage, and the gender income gap. Their connection through female labor supply is

$$\begin{aligned} n &= \mathcal{M}n^m + (1 - \mathcal{M})n^s, \\ l^{\varphi} &= 1 - \chi n, \\ \Gamma^y &= \frac{y^{\sigma}}{y^{\varphi}} = \frac{\Gamma^w}{l^{\varphi}}. \end{aligned} \quad (5)$$

Fertility is measured per woman. Labor incomes are measured before private transfers: $y^{\sigma} = w^{\sigma}$ and $y^{\varphi} = w^{\varphi}l^{\varphi}$.

Marriage is the model's family-structure outcome. The preferred empirical measure is the share of children living with both parents, which depends on both the marriage rate and the ratio of marital to single fertility. The two measures move in the same direction in all individual-policy comparisons and in Proposition 5, because both are increasing in $\mathcal{M}$ and in the relative marital fertility $q = n^m/n^s$.⁹

⁹With relative marital fertility $q = n^m/n^s$, the child-based share equals $\frac{\mathcal{M}q}{1+\mathcal{M}(q-1)}$. This monotonicity gives their comparative statics a common interpretation. Their levels remain distinct, and their movement under other changes

The last relation captures the distinction between wages and incomes. Even at a fixed wage gap, the income gap widens when women spend more time outside the market caring for children. Fertility therefore affects gender income inequality through labor supply, while marriage affects fertility by changing household resources and the composition of families.

3.2 Equilibrium Characterization

Before solving the model, I state the equilibrium formally.

Definition 1 (Competitive equilibrium). *Given primitives $(\beta, \rho, \chi, \lambda, \theta)$ and wages $(w^{\sigma}, w^{\varphi})$, a competitive equilibrium consists of single women's choices, married women's choices, a transfer share α , and marriage rates satisfying:*

1. *Single women optimize subject to their resource and time constraints. Single men remain childless and consume their earnings.*
2. *Wives optimize conditional on the transfer and wages.*
3. *Husbands evaluate marriage at the fertility chosen by wives.*
4. *Taste-adjusted choices generate the participation rates in (3).*
5. *The transfer clears the marriage market as in (4).*

The equilibrium can be solved in closed form. To keep the expressions manageable, define

$$\begin{aligned} r &\equiv \frac{\rho - 1}{\rho}, \\ C &\equiv \left(\frac{(1 - \beta)\chi}{\beta} \right)^{\rho} (w^{\varphi})^{\rho-1} + \chi, \\ K &\equiv 1 - \frac{\beta}{(1 - \beta)(Cw^{\sigma})^r}. \end{aligned} \tag{6}$$

The wife's first-order condition then gives

$$n^s = \frac{1}{C}, \quad n^m = \frac{1 + \alpha\Gamma^w}{C}. \tag{7}$$

Equation (7) makes the income-sharing mechanism transparent. Single and married women face the same monetary price of a child, $w^{\varphi}\chi$. Marriage adds the transfer while leaving that price unchanged. Marital fertility is consequently single fertility multiplied by the increase in resources

depends on fertility composition. Appendix D.4 provides the mapping.

relative to the wife's own earning capacity. If the transfer were zero, the two fertilities would coincide.

The closed-form solution must also respect the household time constraints. I therefore impose the following restrictions.

Assumption 1 (Interior baseline equilibrium). *The primitives satisfy*

$$Cw^{\sigma} > \left(\frac{\beta}{1-\beta} \right)^{1/r}, \quad \frac{\chi(1+\Gamma^w)}{C(1+K^{1/r}\Gamma^w)} < 1. \quad (8)$$

Policy comparisons remain within the region with strictly positive consumption and market labor supply.

The first inequality implies $0 < K < 1$ and delivers strictly positive consumption after the transfer. The second ensures that the associated marital fertility leaves the wife non-negative time for market work. Together, they ensure that the marriage-market solution corresponds to a feasible interior household allocation. Single women's labor supply is already positive because $C > \chi$.

Lemma 1 (Baseline equilibrium). *Under Assumption 1, the interior equilibrium is unique and satisfies*

$$\alpha^* = \frac{1 - K^{1/r}}{1 + K^{1/r}\Gamma^w} \in (0, 1), \quad n^m = \frac{1 + \Gamma^w}{C(1 + K^{1/r}\Gamma^w)}. \quad (9)$$

The transfer and household fertilities are independent of λ and θ .

The proof appears in Appendix D.3. The independence from λ has a useful interpretation. A common proportional increase in the benefit of marriage makes marriage more attractive to both genders. It raises participation while preserving the transfer needed to balance the two sides, because the common benefit cancels from (4). Similarly, θ governs participation's responsiveness to gains, while the equality of those gains determines the equilibrium transfer.

Lemma 2 (Marriage gains). *In the baseline equilibrium,*

$$\begin{aligned} \frac{V_{\varphi,m}^{\varphi}}{V_{\varphi,s}^{\varphi}} &= \lambda(1 + \alpha^*\Gamma^w), & \frac{n^m}{n^s} &= 1 + \alpha^*\Gamma^w > 1, \\ \mathcal{M} &= \frac{[\lambda(1 + \alpha^*\Gamma^w)]^{\theta}}{1 + [\lambda(1 + \alpha^*\Gamma^w)]^{\theta}}. \end{aligned} \quad (10)$$

Lemma 2 states the economic logic compactly. Women's gains from marriage have a common marriage-benefit component, λ , and an income-sharing component, $1 + \alpha^*\Gamma^w$. The latter depends on what the transfer is worth relative to the wife's own earning capacity. Because the price of fertility is unchanged, the same factor scales consumption, fertility, and utility across being single

versus being married. Market clearing equates the spouses' marriage-to-singlehood utility ratios, while their utility levels can differ. Appendix D.2 provides the proof.¹⁰

3.3 The Three-Way Trade-Off

Equations (5) and (10) bring the three outcomes together. Marriage is more attractive when income sharing adds substantially to the wife's resources. A wider wage gap can support that attraction, while also widening the gender income gap. Higher fertility adds a further tension by reducing the female market work that could otherwise offset the wage disparity.

The three possible pairs of objectives reveal different sides of the same problem. Higher fertility leaves women less time for market work, so if marriage is sustained by a substantial male wage advantage and the transfers it supports, the resulting labor-supply difference compounds the gender income gap. When women spend substantial time raising children, income equality requires a relatively small male wage advantage. This leaves less scope for transfers to make marriage attractive relative to women's own earning opportunities. At a given common benefit of marriage, sustaining participation requires an attractive income-sharing component. Reconciling that component with a small income gap is easier when women supply more market labor, which in the baseline requires devoting less time to children.

The scope of this tension depends on the economic environment. The transfer bound makes that dependence precise: every interior baseline equilibrium satisfies

$$\frac{1}{\lambda} \left(\frac{\mathcal{M}}{1 - \mathcal{M}} \right)^{1/\theta} < 1 + \Gamma^y(1 - \chi^n). \quad (11)$$

At fixed λ and χ , sufficiently ambitious joint targets violate (11). Raising the marriage target increases the left side, while raising fertility or lowering the income-gap target reduces the right side. Appendix D.9 derives the restriction.

Rationalizing the cross-country patterns

Before turning to policy, it is useful to ask what the model predicts about the *source* of the shortfall in different economic environments. The empirical decomposition in Section 2.4 reveals that the fertility–gender equality tension dominates in non-OECD countries, while the fertility–dual parenthood tension dominates in the OECD. The model's equilibrium characterization traces this

¹⁰For completeness, Appendix D.1 also writes the husband's participation condition as a transfer schedule conditional on marriage and fertility. That schedule shows how much he is willing to transfer for a given marital bundle. It complements the wife's fertility first-order condition: she chooses fertility, and he decides whether the resulting contract is worth entering.

contrast to a single state variable: the gender wage gap Γ^w . Two mechanisms are at work, one linking fertility to the income gap and the other linking fertility to marriage.

Lemma 3 (Dominant tension and the wage gap). *Consider two economies sharing primitives $(\beta, \rho, \chi, \lambda, \theta)$ and male wage w^σ , with $w^\sigma > w^\varphi > 0$ and interior equilibria, but differing in the gender wage gap $\Gamma^w = w^\sigma/w^\varphi$.*

(i) **F–G amplification.** *The gender income gap satisfies $\Gamma^y = \Gamma^w/(1 - \chi n)$. The maximum fertility consistent with a gender-equality threshold $\bar{\Gamma} > \Gamma^w$ is*

$$\bar{n}(\Gamma^w) = \frac{1}{\chi} \left(1 - \frac{\Gamma^w}{\bar{\Gamma}} \right),$$

which is strictly decreasing in Γ^w and equals zero when $\Gamma^w = \bar{\Gamma}$.

(ii) **F–M coupling.** *The marital fertility premium $q - 1 = (1 - g)\Gamma^w/(1 + g\Gamma^w)$, where $g = K^{1/r} \in (0, 1)$, is strictly increasing in Γ^w through two reinforcing channels: the direct effect of a larger wage gap and the indirect effect through a lower g . The composition elasticity of aggregate fertility with respect to marriage, $\varepsilon_{n,\mathcal{M}} = \mathcal{M}(q - 1)/[1 + \mathcal{M}(q - 1)]$, is also increasing in Γ^w .*

The proof appears in Appendix D.5. The two parts of the lemma work together to generate the empirical pattern. In non-OECD countries, where gender wage gaps are large, part (i) implies that the fertility ceiling compatible with gender equality is low: even moderate fertility pushes the income gap above the equality threshold. This is the F–G tension. Part (ii) implies that the marital fertility premium is large, so marriage and fertility are tightly co-determined through the income-sharing channel. Cross-country variation in primitives moves marriage and fertility in the same direction, making the F–M tension unlikely. The model therefore predicts that at high Γ^w the F–G tension dominates and the F–M tension is weak, matching the non-OECD decomposition in Table 4 (55.9 versus 26.7 percent of the shortfall).

In OECD countries, where gender wage gaps are narrower, the picture reverses. Part (i) implies that the fertility ceiling is high, so moderate fertility is compatible with gender equality and the F–G tension largely disappears.¹¹ Part (ii) implies that the marital fertility premium is small ($q \approx 1$), weakening the composition channel that ties marriage to fertility. Variation in marriage-relevant parameters, such as social norms and cohabitation institutions, then generates marriage differences across countries without proportional fertility differences, while variation in childcare

¹¹This is the pattern emphasized by Doepke et al. (2023), who attribute the growing compatibility of fertility and gender equality in advanced economies to a broad set of institutional adaptations. The model here complements their account by identifying the wage gap as a sufficient statistic for the F–G tension: as Γ^w narrows, fertility’s amplification of income inequality weakens, regardless of which specific institutional channels drive the narrowing.

costs generates fertility differences without proportional marriage differences. This decoupling allows the F–M tension to emerge as the dominant constraint, matching the OECD decomposition (63.4 versus –1.1 percent of the shortfall).

The M–G tension is present at both levels of Γ^w . A higher marriage rate raises aggregate fertility through composition, reducing female market work and widening the income gap. This mechanism operates regardless of the initial wage gap, consistent with the moderate M–G contributions in both subsamples (45.1 and 37.9 percent).

Appendix H asks whether alternative candidates—wage levels, fertility preferences, or the childcare division—could independently generate these contrasting patterns, and finds that Γ^w is uniquely positioned, entering both the F–G and F–M mechanisms through direct channels while no single alternative does.

The policy question is now how the available instruments move the three outcomes within this economic environment. The next subsection establishes why the conventional instruments considered individually leave the quandary unresolved, and then asks whether combining them can do better.

3.4 Policy Remedies

I consider three familiar policy objectives in turn: making marriage more attractive, improving women’s wage opportunities, and reducing the time cost of children. Each experiment changes one primitive while holding the others fixed. These experiments isolate mechanisms. Applying them to a complete fiscal program would additionally require specifying financing and the other features of the reform.

Pro-marriage policies

Marriage bonuses, joint taxation, and related measures seek to make marriage more attractive, partly because of its connection to family stability and children’s outcomes (Kearney, 2023; Autor et al., 2019). I represent their common-benefit component by an increase in λ .

Proposition 1. *An increase in λ raises aggregate fertility n and marriage M but widens the gender income gap Γ^y .*

The logic is compositional. A higher λ attracts more people into marriage while leaving the equilibrium transfer and fertility within each marital state unchanged. Since married women have more children than single women, average fertility rises as more women marry. Those additional children require maternal time. Female labor supply falls, and the income gap widens. The policy succeeds on marriage and fertility through a channel that works against income equality. Appendix D.6 proves the result.

Gender equality policies

Anti-discrimination and pay-equity measures seek to improve women's labor-market opportunities and reduce misallocation (Hsieh et al., 2019). I isolate their wage effect through an increase in w^{f} at fixed w^{m} .

Proposition 2. *An increase in w^{f} , holding w^{m} fixed, reduces aggregate fertility n and marriage $\mathcal{M}$ while narrowing the gender income gap Γ^y .*

Higher female wages act through two channels. First, children become more expensive in terms of forgone earnings, reducing fertility among both single and married women. Second, a husband's transfer adds less relative to the wife's own earning capacity, reducing the income-sharing attraction of marriage. Fewer marriages then reinforce the fertility decline.

The transfer share responds to competing forces. Wage convergence and the changing value of fertility exert opposing effects on α^* . Their combined effect unambiguously reduces the transfer relative to women's own earning capacity, $\alpha^*\Gamma^w$, and therefore the common marriage premium. Meanwhile, women both earn more per unit of work and spend more time in the market, so the income gap narrows through wages and labor supply together. Appendix D.7 establishes these claims.

Reductions in childcare time costs

A lower time requirement per child appears to offer a direct way to support fertility while preserving women's market work. To examine that possibility, I reduce χ while holding wages and the remaining primitives fixed.

Proposition 3. *A reduction in χ raises aggregate fertility n and marriage $\mathcal{M}$ but widens the gender income gap Γ^y .*

Lower childcare costs raise fertility in both marital states. They also increase the equilibrium value of the marital bundle, supporting a larger transfer and more marriage. These direct and composition effects both raise aggregate fertility.

Why, then, does female labor supply fall? The answer lies in the distinction between care per child and total care. With $\rho > 1$, families respond sufficiently strongly to the lower time price that χn rises even though χ falls. Mothers spend less time on each child but more time on childcare altogether. The resulting reduction in market work widens the income gap. Appendix D.8 establishes the response of total childcare time.

Three instruments, three incomplete solutions

Table 5 summarizes the results. Each conventional instrument succeeds on part of the agenda while leaving at least one objective moving in the wrong direction. Pro-marriage and lower-childcare-cost policies raise fertility and marriage at the expense of income equality. Higher female wages narrow the income gap but reduce fertility and marriage.

Policy	Fertility n	Marriage $\mathcal{M}$	Income gap Γ^y
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\text{f}}$)	−	−	− (narrows)
Lower childcare costs ($\downarrow \chi$)	+	+	+ (widens)

Table 5. Effects of Conventional Policies Considered Individually

Notes: Each row changes one instrument while holding the remaining primitives fixed. Comparisons remain within the interior equilibrium region.

The pattern in Table 5 raises a natural question: can coordinating instruments overcome the limitations of each individual one?

Coordinating instruments

The key observation is that the pro-marriage benefit λ and female wages w^{f} have opposite effects on fertility and marriage, but the same desirable effect on income equality (through different channels). By combining a wage increase with a marriage-benefit increase, one can use the latter to offset the former's adverse effects on fertility and marriage while preserving the income-gap gains. The following proposition makes this precise.

Proposition 4 (Coordinated instruments). *At an interior baseline equilibrium, a sufficiently small increase in w^{f} combined with an appropriately chosen increase in λ raises aggregate fertility n and marriage $\mathcal{M}$ while narrowing the gender income gap Γ^y . The child-based family-structure counterpart also improves.*

The construction works as follows. Start with a small increase in w^{f} , which by Proposition 2 reduces fertility and narrows the income gap. Simultaneously raise λ just enough to hold aggregate fertility constant. Because household fertilities and the transfer are independent of λ (Lemma 1), the constant-fertility condition requires marriage to rise: the decline in both n^m and n^s means that the composition must shift toward the higher-fertility marital state. Female labor supply, $l^{\text{f}} = 1 - \chi n$, is unchanged along this path, so the income gap narrows purely through the higher female wage. A slightly larger marriage-benefit increase then raises fertility strictly, and continuity preserves the strict marriage increase and income-gap decline. The child-based family-structure measure, which

equals $1 - (1 - \mathcal{M})n^s/n$ at constant aggregate fertility, also rises because marriage increases and single fertility falls. The proof appears in Appendix D.9.

The result establishes that the quandary is not an inherent feature of the economic environment. At any interior equilibrium, regardless of initial wages, preferences, or the existing level of gender inequality, an appropriately coordinated policy mix can move all three outcomes in the desired direction. This stands in contrast to the redistribution result that follows, which requires additional conditions.

The practical limitation is that the required marriage-benefit adjustment must be calibrated precisely to the wage change. Too small an adjustment leaves fertility below its initial level; too large an adjustment reverses the income-gap gains. Such fine-tuning places demanding informational requirements on the policymaker and requires that two distinct policy levers (labor-market interventions and marriage incentives) be adjusted in tandem. These difficulties motivate the search for a simpler approach: can a single policy instrument achieve the same goal without requiring such delicate coordination? The next section asks whether redistributing childcare between spouses provides such an instrument.

4 Redistributing Childcare Between Spouses

The baseline assigns all childcare to mothers. Time-use evidence documents a substantial maternal share of care (Aguiar and Hurst, 2007; Kleven et al., 2019), and institutions and policies, including reserved paternity leave, can change fathers' involvement (Farré and González, 2019). This raises a natural question: can sharing care more equally support fertility and marriage while maintaining or improving gender income equality?

Answering that question requires following the time savings through both spouses' budgets. When the husband provides additional care, he earns less, and some of that income would have financed his transfer to the wife. He must also be willing to accept the new balance between consumption and children. To capture these effects, I extend the model to allow both spouses to provide childcare and let the transfer adjust to clear the marriage market.

4.1 Setup and Household Choices

Let $\chi^{\varnothing}$ and χ^{σ} denote the *effective* childcare per child provided by wives and husbands:

$$\chi^{\varnothing} > \chi^{\sigma} \geq 0, \quad \chi^{\varnothing} + \chi^{\sigma} = \chi. \quad (12)$$

Single mothers continue to provide χ units of care per child. Marriage thus changes who provides the care while preserving the total effective parental time required for a child. Appendix E.7 gives

the corresponding analysis when total requirements differ by marital status.

A parameter $\eta \in (0, 1]$ captures *paternal care efficiency*: one unit of the father's time produces η effective units of childcare, so his raw time per child is χ^{σ} / η . When $\eta = 1$ both parents are equally efficient; when $\eta < 1$ the father requires more time to deliver the same effective care. A large empirical literature documents that mothers tend to be more practiced at caregiving and that fathers' involvement rises with experience (Farré and González, 2019; Patnaik, 2019), making $\eta \leq 1$ the empirically relevant range.

The wife's budget becomes

$$c^{\varnothing, m} = \alpha w^{\sigma} \left(1 - \frac{\chi^{\sigma}}{\eta} n^m\right) + w^{\varnothing} (1 - \chi^{\varnothing} n^m),$$

and the husband's budget becomes

$$c^{\sigma, m} = (1 - \alpha) w^{\sigma} \left(1 - \frac{\chi^{\sigma}}{\eta} n^m\right).$$

The transfer remains a share of the husband's earnings, which now depend on his childcare obligations. As before, the wife chooses fertility conditional on the transfer share, and the husband evaluates the resulting bundle.

It is useful to rewrite the wife's budget in terms of effective resources and an effective childcare cost:

$$\tilde{w}^{\varnothing} = w^{\varnothing} + \alpha w^{\sigma}, \quad \tilde{\chi}^{\varnothing} = \frac{w^{\varnothing} \chi^{\varnothing} + \alpha w^{\sigma} \chi^{\sigma} / \eta}{w^{\varnothing} + \alpha w^{\sigma}}. \quad (13)$$

Her budget is then $c^{\varnothing, m} = \tilde{w}^{\varnothing} (1 - \tilde{\chi}^{\varnothing} n^m)$. The effective childcare cost is an income-weighted average: a child uses some of her own earning capacity and some of the husband's, the latter deflated by η because the father requires $1/\eta$ raw hours per effective unit of care.

The interior fertility choices are

$$n^m = \frac{1}{\left(\frac{(1-\beta)\tilde{\chi}^{\varnothing}}{\beta}\right)^{\rho} (\tilde{w}^{\varnothing})^{\rho-1} + \tilde{\chi}^{\varnothing}}, \quad n^s = \frac{1}{C}. \quad (14)$$

This is the same household trade-off as before, expressed using the resources and fertility price relevant within marriage.

Lemma 4 (Indirect utility with shared childcare). *The wife's marital value is*

$$V^{\varnothing, m} = \lambda (1 - \beta)^{1/r} \tilde{w}^{\varnothing} \left[1 + \left(\frac{\beta}{1 - \beta} \right)^{\rho} (\tilde{w}^{\varnothing} \tilde{\chi}^{\varnothing})^{1-\rho} \right]^{1/(\rho-1)}. \quad (15)$$

Her single value follows by setting $\lambda = 1$, $\tilde{w}^{\varnothing} = w^{\varnothing}$, and $\tilde{\chi}^{\varnothing} = \chi$.

Equations (14) and (15) show why the effective price matters. At fixed resources, a lower price raises fertility and the value of marriage to the wife. With $\rho > 1$, she substitutes sufficiently strongly toward children that consumption falls. Her utility gain comes from the opportunity to choose a more attractive combination of consumption and fertility. Appendix E.1 provides the derivation.

4.2 When Redistribution Improves All Three Outcomes

Consider a local policy that shifts one *effective* unit of childcare per child from the wife to the husband:

$$\frac{d\chi^{\sigma}}{d\delta} = 1, \quad \frac{d\chi^{\varnothing}}{d\delta} = -1.$$

Because the father's raw time per child is χ^{σ}/η , this shift requires $1/\eta$ additional raw hours from him. When $\eta = 1$ the experiment coincides with a one-for-one time swap; when $\eta < 1$ the father devotes more raw time than the mother saves, reflecting his lower efficiency.

Wages, preferences, and single mothers' childcare costs remain fixed. Total derivatives below include the equilibrium adjustment of α ; at $\chi^{\sigma} = 0$, they are interpreted as right-hand derivatives. Appendix E.2 derives the local equilibrium response formulas used below.

The experiment treats the childcare division as a policy variable while holding wages and preferences fixed. In practice, these objects change at different speeds. Over the sample period in Mexico, the relative gender wage gap declined from 2.9 to 1.8, while the female childcare share fell only from 0.85 to 0.73. Deschamps et al. (2026) document a similar pattern globally: child penalties in time use vary enormously across countries at similar income levels, with the cross-country variation tracking cultural proxies more closely than economic fundamentals. The slow adjustment of childcare norms relative to market wages (Goldin, 2025) motivates treating the division as a lever that shifts exogenously, whether through mandated paternity leave, workplace regulation, or gradual norm change (Albrecht et al., 2024; Xi and Zhou, 2026).

Three sufficient conditions organize the result. The first and second concern the female side of the reform: does it make fertility less costly for the wife, and does it improve women's earnings relative to men's after fertility adjusts? The third asks whether the husband is willing to enter the modified contract. Together, they follow the change in caregiving through household choices, marriage participation, and relative earnings.

Condition 1: the wife faces a lower effective childcare cost. At the initial transfer, redistribution lowers the wife's effective childcare cost if

$$w^{\varphi} > \frac{\alpha w^{\sigma}}{\eta}. \quad (16)$$

The intuition is straightforward. An effective hour released from the wife's childcare raises her potential earnings by w^{φ} . Replacing that effective hour requires $1/\eta$ raw hours of the husband's time, reducing the earnings available to finance her transfer by $\alpha w^{\sigma}/\eta$. If the first exceeds the second, shifting care toward the husband lowers the price of children from her perspective. Her desired fertility and marital value both rise at the initial transfer.

Condition (16) compares the earning capacities affected by the redistribution, accounting for the father's efficiency. When $\eta = 1$ it reduces to $w^{\varphi} > \alpha w^{\sigma}$; lower η makes the condition harder to satisfy because each effective unit of paternal care consumes more of the father's market time and hence more of the transfer base.

Condition 2: relative earnings converge after fertility adjusts. The second condition follows the reform through aggregate childcare. If redistribution raises fertility and marriage, it also creates additional care to be provided. Income convergence depends on how that larger total is divided between mothers and fathers.

Average labor supplies and the income ratio are

$$\begin{aligned} l^{\sigma} &= 1 - \frac{\chi^{\sigma}}{\eta} \mathcal{M}n^m, \\ l^{\varphi} &= 1 - \chi n + \chi^{\sigma} \mathcal{M}n^m, \\ \Gamma^y &= \Gamma^w \frac{l^{\sigma}}{l^{\varphi}}. \end{aligned} \quad (17)$$

The father's raw time per child, χ^{σ}/η , determines male labor supply. The wife's effective care contribution χ^{σ} enters female labor supply because each effective unit the father provides is an effective unit the mother does not. The income-gap condition is

$$\chi \frac{dn}{d\delta} < \left[\mathcal{M}n^m + \chi^{\sigma} \frac{d(\mathcal{M}n^m)}{d\delta} \right] \left(1 + \frac{l^{\varphi}}{\eta l^{\sigma}} \right). \quad (18)$$

The left side is the increase in total childcare generated by higher fertility. The bracket on the right is the increase in fathers' aggregate *effective* childcare, including their care for additional marital births. The final factor accounts for the two-sided effect on relative earnings: each effective

unit the father takes on costs him $1/\eta$ raw hours of market time while freeing one hour for the mother. When $\eta < 1$ this asymmetry amplifies the male-side adjustment, making the condition easier to satisfy.

Condition (18) requires women's earnings to improve relative to men's. This can occur even if the fertility response reduces female labor supply, provided male labor supply falls proportionately more.

Condition 3: the husband gains from the adjusted marital contract. The equilibrium response also depends on the husband's valuation of the reform. He loses earnings when he provides more care, but he may value the additional fertility enough to accept that loss. Moreover, the transfer can adjust to redistribute the gains. To express this trade-off, define

$$z \equiv \tilde{\chi}^{\varnothing} n^m \in (0, 1), \quad R \equiv \frac{\beta}{1 - \beta} \left(\frac{c^{\sigma, m}}{n^m} \right)^{1/\rho} - (1 - \alpha) w^{\sigma} \frac{\chi^{\sigma}}{\eta}.$$

Here z is the share of the wife's effective resources devoted to fertility. The term R is the husband's marginal value of another child, measured in consumption units and net of the earnings cost of his existing care obligation—the latter scaled by $1/\eta$ because each effective unit of care requires $1/\eta$ raw hours.

The male-side condition is

$$\frac{R \left(w^{\varnothing} - \frac{\alpha w^{\sigma}}{\eta} \right) \rho (1 - z)}{\tilde{w}^{\varnothing} \tilde{\chi}^{\varnothing}} > \left(1 - \frac{\chi^{\sigma}}{\eta} n^m \right) \left(\frac{w^{\sigma}}{\eta} - w^{\varnothing} \right). \quad (19)$$

A compensated comparison gives (19) its economic meaning. Imagine adjusting the transfer as childcare is reallocated so that the wife's gain from marriage stays at its original level. Fertility responds to both the lower childcare cost and this transfer adjustment. The condition requires the husband to gain along that path. After ensuring that the wife is no worse off, the additional children must be valuable enough to leave a gain for him as well.

The left side measures the husband's net valuation of that induced fertility response. The right side captures the raw-time cost of shifting care toward the higher-wage spouse: even after the compensating transfer, the father's $1/\eta$ raw hours per effective unit command a wage of w^{σ} , while the mother's freed hour commands $w^{\varnothing}$. The gap $w^{\sigma}/\eta - w^{\varnothing}$ is the per-unit cost of the efficiency wedge. If the inequality holds, the reform creates room to make marriage more attractive to both genders. The actual market-clearing transfer then shares that improvement, raising both spouses' marriage premiums. Appendix E.3 derives this interpretation.

Proposition 5 (Redistributing childcare). *Consider an interior equilibrium with $w^{\sigma} > w^{\varphi} > 0$, $0 < \alpha < 1$, $\eta \in (0, 1]$, and (12). If Conditions (16) and (19) hold, the equilibrium is locally regular and a sufficiently small redistribution toward fathers raises marriage, marital fertility, and aggregate fertility. Both spouses' marriage-to-singlehood utility ratios rise. If Condition (18) also holds, the gender income gap narrows. Thus the three conditions together imply*

$$\frac{dn}{d\delta} > 0, \quad \frac{d\mathcal{M}}{d\delta} > 0, \quad \frac{d\Gamma^y}{d\delta} < 0.$$

The proof appears in Appendix E.4. The result works through household choices and market participation together. The lower effective price encourages fertility, while the husband's gain from the adjusted bundle supports a higher common marriage premium. More people marry, and married women choose more children. Because single fertility is unchanged and marital fertility is higher, both changes raise aggregate fertility.

Income convergence requires a further step. Additional children bring additional care obligations, and redistribution must be sufficient to improve women's earnings relative to men's after those obligations are taken into account. Conditions (16) and (19) resolve the participation problem; Condition (18) resolves the relative-income problem.

Starting from the baseline, $\chi^{\sigma} = 0$, the relative-earnings condition simplifies to

$$\chi \frac{dn}{d\delta} < \frac{\mathcal{M}n^m}{\eta} (\eta + l^{\varphi}).$$

Appendix E.5 gives the corresponding response formulas using the closed-form baseline equilibrium.

The minimum efficiency threshold. All three conditions become harder to satisfy as η falls, because each effective unit of paternal care requires progressively more raw time from the higher-wage father.¹² For a given economic environment $(\alpha, w^{\sigma}, w^{\varphi}, n^m, z, R)$, define the *minimum paternal efficiency*

$$\bar{\eta} \equiv \min\{\eta \in (0, 1] : \text{Conditions (16), (18), and (19) all hold}\}. \quad (20)$$

When $\eta \geq \bar{\eta}$, Proposition 5 applies and redistribution can improve all three outcomes; when $\eta < \bar{\eta}$, at least one condition fails. A lower $\bar{\eta}$ indicates a wider scope for redistribution, because fathers can be less efficient and the result still goes through. As wages converge (Γ^w falls), the earnings cost of paternal care decreases, typically reducing $\bar{\eta}$. The quantitative section evaluates this threshold

¹²Condition (18) is the exception: lower η amplifies the male labor-supply cost of care, which favours income convergence. In practice, however, the overall threshold is governed by Condition (19).

for Mexico.

Propositions 4 and 5 offer complementary approaches to the quandary. Coordinated instruments always work at any interior equilibrium but require two policy levers calibrated in tandem. Redistribution uses a single lever but requires three conditions: on the wife's effective price, the husband's willingness to participate, and relative earnings after fertility adjusts. When those conditions hold, redistribution provides a simpler path to the same goal. The broader lesson is bilateral: mothers' cost of children and fathers' valuation of the resulting marital bundle jointly determine how fertility responds. Redistribution can improve all three outcomes when it creates an attractive exchange for both sides of the marriage market and leaves relative earnings more equal after families adjust.

4.3 When the Conditions Fail

The same framework clarifies why a reform can help in one setting and disappoint in another. For example, the equilibrium transfer response can offset a direct reduction in the wife's childcare cost. Alternatively, a contract that raises both spouses' marriage gains can also generate enough additional care to prevent income convergence. These are distinct ways for joint improvement to fail.

For the following proposition, maintain $w^{\sigma} > w^{\varphi} > 0$, $0 < \alpha < 1$, $\eta \in (0, 1]$, interior household allocations, and the fixed-total-care experiment. Consider a regular equilibrium branch on which

$$\frac{\partial}{\partial \alpha} \log \left[\frac{V^{\varphi,m}/V^{\varphi,s}}{V^{\sigma,m}/V^{\sigma,s}} \right] > 0.$$

This condition says that increasing the transfer raises women's willingness to marry relative to men's, including the induced fertility response. Conditions (16) and (19) imply it.

Proposition 6 (Alternative responses). *At an equilibrium satisfying the preceding regularity condition:*

- (A) *If $w^{\varphi} \leq \alpha w^{\sigma} / \eta$, the wife's fertility and marital utility weakly fall at the initial transfer. Starting from $\chi^{\sigma} = 0$, marriage, marital fertility, and aggregate fertility fall in equilibrium, while the gender income gap narrows.*
- (B) *If $w^{\varphi} > \alpha w^{\sigma} / \eta$ and the husband's marital utility weakly falls at the initial transfer, the equilibrium transfer falls. The equilibrium marriage and fertility responses depend on the transfer-adjusted incentives.*
- (C) *If Conditions (16) and (19) hold but Condition (18) fails, marriage and fertility rise, while the gender income gap is locally unchanged or widens.*

Appendix E.6 provides the proof. Case (A) reverses the wife’s price mechanism. When her budget depends sufficiently heavily on her husband’s earnings—or when paternal efficiency is low enough that each effective unit of care consumes a large share of the transfer base—shifting care toward him removes more transfer-financing capacity than it releases own earning capacity. At the baseline, marriage and fertility decline. With positive paternal care initially, additional interactions arise because lower fertility also reduces the husband’s existing childcare burden.

Case (B) illustrates the importance of the transfer adjustment. At the original transfer, the wife gains and the husband’s utility weakly falls. The transfer must fall to restore balance. That adjustment returns some resources to the husband while changing the wife’s fertility choice, so the equilibrium fertility response can differ from its initial direction.

Case (C) concerns aggregate time allocation. The marriage-market response is favorable: both sides value the contract more, and fertility rises. The additional childcare is sufficiently large, however, that the per-child redistribution fails to narrow the aggregate income gap. Relative earnings depend on the division of care across all children as well as the division per child.

The relative strength of these mechanisms depends on equilibrium quantities that change with wages, preferences, and childcare arrangements. Their implications for high-income or middle-income countries therefore require evaluation at each economy’s starting point, which calls for careful estimation of parameters.

4.4 Relationship to the Empirical Evidence

Reserved paternity leave provides a natural setting in which to study changes in fathers’ caregiving. Evidence from Spain (Farré and González, 2019), Norway (Cools et al., 2015), Denmark (Kleven et al., 2026), South Korea (Lee and Lee, 2025), and Quebec (Patnaik, 2019) examines how reforms change the household division of labor. These outcomes address the first practical question: does policy move childcare toward fathers? The literature generally finds a positive answer.

The next question is whether that change supports fertility and marriage. The literature reports differing fertility responses across the settings studied. For example, Farré and González (2019) finds a reduction in subsequent fertility while Lee and Lee (2025) finds an increase in fertility following paternal leave policies. The model offers a way to organize this heterogeneity. A reform changes mothers’ time constraints alongside fathers’ consumption, private transfers, and the attraction of the marital bundle. Similar shifts in caregiving can therefore produce different fertility responses depending on preferences and initial conditions. This calls for a case-by-case calibration à la Section 5.

The third question is whether the gender income gap narrows after families respond. Higher maternal employment is one route to income convergence. The relative-income measure used in this paper also captures changes on the male side, allowing convergence when men’s earnings fall

proportionately more. Evaluating the full prediction therefore calls for information on wages, labor supply, fertility by marital status, private transfers, and marriage participation.

4.5 Public Childcare and the Marketization of Childcare

Public childcare and purchased care provide another way to release parental time. Outsourcing can reduce total parental care per child, introduce a monetary payment, and change the two parents' remaining obligations differently. Its effects therefore combine several mechanisms.

The baseline reduction in χ isolates a lower maternal time requirement while holding additional spending and financing outside the experiment. Proposition 3 shows that, under $\rho > 1$, the fertility response is strong enough for total maternal childcare to rise. This result highlights the distinction between time saved per child and total time saved after fertility adjusts. Applying it to public childcare would require accounting for the reform's other effects on household resources and choices.

The redistribution experiment focuses on who provides the remaining care. Moving an effective hour from the wife to the husband lowers his earnings by w^σ/η as it releases w^φ of hers. Purchased childcare replaces parental time with a service whose cost must be financed. The associated budgets and participation incentives differ, so extending Proposition 5 to outsourced care requires incorporating those payments and their incidence.

The common lesson is to follow the reform through the household and the marriage market. How does it change the wife's effective price of children? What contract is the husband willing to accept after fertility and transfers adjust? And how is total childcare ultimately divided between parents and the market? A model with purchased childcare would answer these questions by specifying prices, financing, and substitution with parental time.

5 Quantitative Assessment

I calibrate the extended model, in which both spouses provide childcare, to Mexico from 1990 to 2025. The exercise exploits a long time series, substantial productivity and wage changes during the demographic transition, the availability of micro-level data on the gender division of childcare, and a trajectory broadly representative of transitions in many middle-income economies.

5.1 Data

Core variables

The calibration draws on three categories of data. *Fertility* is the total fertility rate from the United Nations. *Dual parenthood* measures children's co-residence with both parents, from [Brenøe and](#)

Wasserman (2025).¹³ The *female labor income share* is from the World Inequality Database.

Additional data

Two additional data series, not used in the cross-country analysis, are required for the extended-model calibration.

GDP per capita in constant 2010 U.S. dollars is from the Federal Reserve Economic Data (FRED). Over 1990–2025, Mexican GDP per capita rises from roughly \$8,000 to approximately \$10,300, with substantial volatility around the 1994–95 Peso Crisis, the 2008–09 recession, and the 2020 pandemic downturn.

The gender division of childcare is constructed from micro-level data from Mexico’s National Survey on Time Use (Encuesta Nacional sobre Uso del Tiempo, ENUT). Three survey waves provide anchor points for the female share of total parental childcare time:

- 1996 (ENTAUT): female share = 0.85;
- 2009 (ENUT, Cuadro 3.1.1): female share = 0.80;
- 2024 (ENUT): female share = 0.73.

A piecewise-linear interpolation through the three anchors generates an annual series. The resulting path declines at approximately -0.38 percentage points per year before 2009 and -0.47 percentage points per year thereafter, reflecting the gradual shift of childcare responsibilities toward fathers documented in the time-use literature.

Figure 4 plots the five series used in the calibration.

5.2 Calibration

Fertility falls from approximately 3.4 to 1.8 children per woman. The share of children living with both parents declines from 81 to approximately 78 percent, with a dip below 75 percent around 2017. The female income share rises from 22 to 34 percent. Real GDP per capita increases from roughly \$8,000 to over \$10,000 in 2010 dollars.

¹³The model’s “marriage” margin should be understood as any co-resident two-parent arrangement, not exclusively legal marriage. The calibration target, children’s co-residence with both biological parents, captures both formal marriages and consensual unions (*uniones libres*), which are widespread in Mexico and grew over the sample period. Because the model’s income-sharing and childcare-sharing mechanisms apply to any stable co-resident union, this broader notion of partnership is the appropriate empirical counterpart. The measured decline in dual parenthood therefore reflects an increase in single-parent households rather than a substitution between legal marriage and cohabitation.

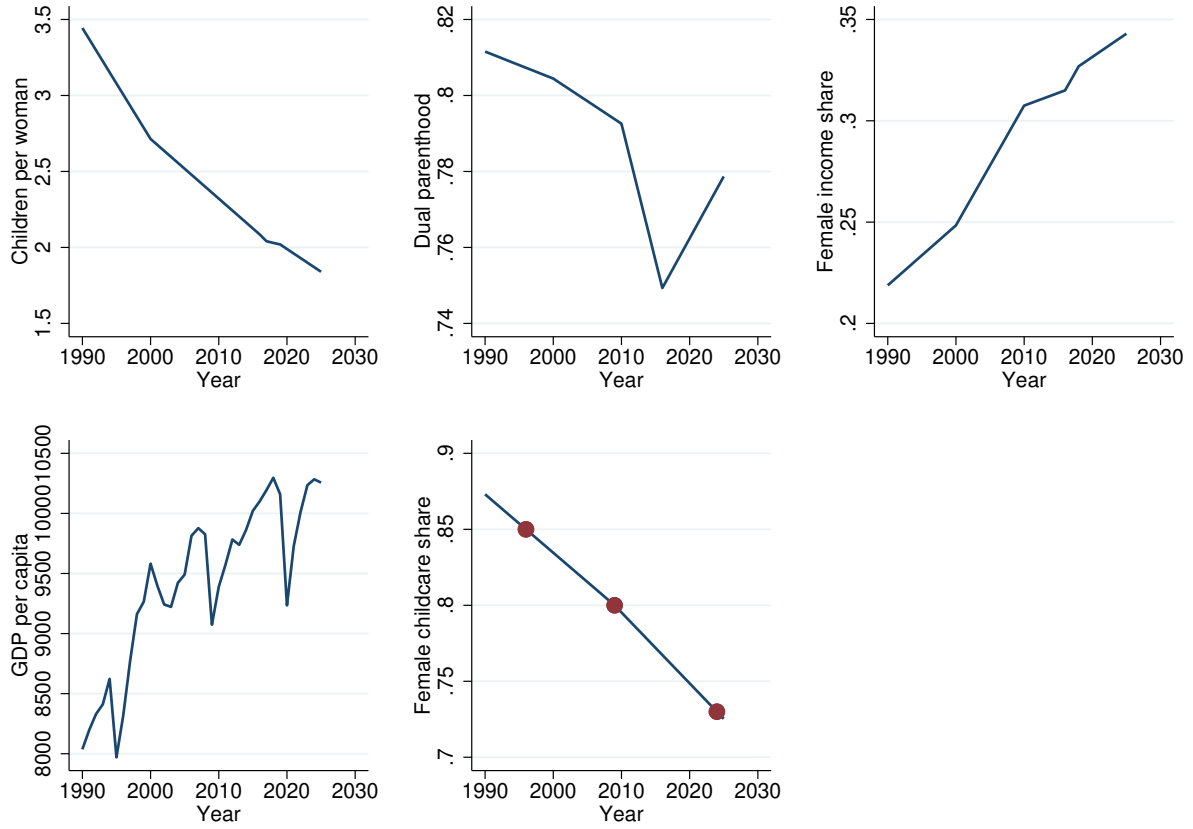


Figure 4. Mexico's Transition Path (1990–2025)

Notes: Children per woman from the United Nations. Dual parenthood measures children's co-residence with both parents (Brenøe and Wasserman, 2025). Female labor income shares from the World Inequality Database. GDP per capita in constant 2010 U.S. dollars from FRED. Female childcare share from a piecewise-linear interpolation through ENUT survey anchors (dots) in 1996, 2009, and 2024.

Drivers

The extended model requires three time-varying inputs in each year t :

1. **Gender-neutral productivity**, A_t , which equals the male wage w_t^{σ} .
2. **The relative wage gap**, $B_t \equiv w_t^{\sigma} / w_t^{\varnothing}$, which captures all gender-biased factors including human capital differences, discrimination, and occupational sorting.
3. **The gender division of childcare**, $(\chi_t^{\varnothing}, \chi_t^{\sigma})$, with $\chi_t^{\varnothing} + \chi_t^{\sigma} = \chi$. The female share is set from the ENUT-based series, so $\chi_t^{\varnothing} = s_t \chi$ and $\chi_t^{\sigma} = (1 - s_t) \chi$, where s_t is the female childcare share.

Throughout the calibration, paternal care efficiency is set to $\eta = 1$, so that one hour of the father's time produces one effective hour of childcare. Section 5.4 examines robustness to $\eta < 1$.

Recovering wages

I set $\chi = 0.075$ following [De La Croix and Doepke \(2003\)](#).¹⁴ Female and male labor supplies are inferred from observed fertility and the childcare-share series:

$$l_t^{\text{f}} = 1 - s_t \chi n_t, \quad l_t^{\text{m}} = 1 - (1 - s_t) \chi n_t,$$

where n_t is observed fertility. Wages are then recovered from GDP per capita and female income shares using the accounting identities

$$w_t^{\text{m}} = \frac{2\tilde{A}_t(1 - \sigma_t)}{l_t^{\text{m}}}, \quad w_t^{\text{f}} = \frac{2\tilde{A}_t\sigma_t}{l_t^{\text{f}}},$$

where $\tilde{A}_t$ is GDP per capita normalized to its 1990 value and σ_t is the female income share. The resulting $A_t = w_t^{\text{m}}$ and $B_t = w_t^{\text{m}}/w_t^{\text{f}}$ are the model's productivity inputs.

Preference and marriage parameters

Four parameters $\{\beta, \rho, \lambda, \theta\}$ are chosen to match the fertility and dual-parenthood series over 1990–2025. The model's marriage rate and relative fertility are mapped into the child-based family-structure measure as in Appendix D.4. The parameters govern distinct margins: β primarily controls fertility's level, ρ its response to productivity changes, λ the level of marriage and thus the mapped family-structure measure, and θ participation responsiveness. Table 6 reports the estimates.

	Interpretation	Value	Target	Source
χ	Time cost of children	0.075	De La Croix and Doepke (2003)	
η	Paternal care efficiency	1	Baseline (robustness in Section 5.4)	
A_t	Gender-neutral productivity		GDP per capita	FRED
B_t	Relative wage gap ($w_t^{\text{m}}/w_t^{\text{f}}$)		Female labor income share	World Inequality Database
s_t	Female childcare share		ENUT anchors	ENUT 1996, 2009, 2024
β	Fertility preference weight	0.077	Fertility level	United Nations
ρ	Consumption–fertility substitutability	2.50	Fertility trend	United Nations
λ	Common marriage benefit	0.954	Dual-parenthood level	Brenøe and Wasserman (2025)
θ	Participation responsiveness	12.0	Dual-parenthood trend	Brenøe and Wasserman (2025)

Table 6. Calibration Results

Notes: Time-varying inputs are constructed from data. Preference and marriage parameters are chosen to match the model's fertility and children's co-residence with both parents to the data over 1990–2025. The gender division of childcare is pinned by three ENUT survey waves.

The estimate $\rho = 2.50$ places the economy firmly in the substitution-dominant region ($\rho > 1$). The estimated marriage benefit $\lambda = 0.954$ is below unity, reflecting that the common marriage-

¹⁴Appendix G re-does the entire calibration exercise for $\chi \in \{0.05, 0.10\}$ and shows that the main results are robust.

benefit shifter combines with the income-sharing component to generate the observed marriage rates. With high participation responsiveness ($\theta = 12$), even modest changes in the marriage-to-singlehood utility ratio produce substantial movements in marriage rates.

To assess local identification formally, Appendix C computes the elasticity Jacobian of the four targeted moments with respect to $(\beta, \rho, \lambda, \theta)$ and its inverse. The forward Jacobian confirms that each parameter has its strongest effect on the intended target moment; the inverse reveals which moments carry the most information about each parameter.

Model fit

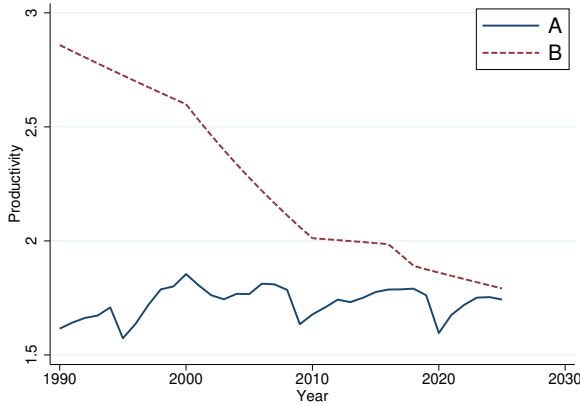
Figure 5 displays the calibrated productivity inputs and the model fit. Over the sample, A_t (gender-neutral productivity) fluctuates modestly around 1.7, while B_t (the relative wage gap) declines sharply from approximately 2.9 to 1.8, reflecting the substantial convergence of female wages toward male wages. The model tracks the observed fertility decline with a correlation of 0.94 and a root-mean-square error of 0.16 children per woman. The dual-parenthood fit has a correlation of 0.52 and an RMSE of 0.02, capturing the level and overall downward trend but less of the short-run variation.

The calibrated wage-gap trajectory also provides a concrete illustration of the mechanism in Lemma 3. Mexico begins the sample at $B_t \approx 2.9$, squarely in the high- Γ^w regime where the model predicts a dominant fertility–gender equality tension. By 2025, B_t has fallen to approximately 1.8, moving the economy toward the lower- Γ^w regime where the fertility–marriage tension becomes relatively more important. The declining wage gap thus shifts the binding constraint within the trinity over the course of the transition.

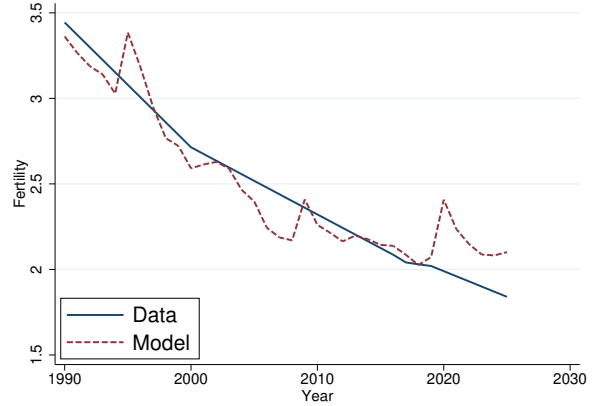
Out-of-sample validation: female labor force participation

The calibration targets fertility and dual parenthood. Female labor supply, $l_t^\varnothing = 1 - \chi_{st}n_t$, is an equilibrium outcome that the model predicts but does not use in estimation. An informative out-of-sample check compares this prediction to observed female labor force participation (FLFP) in Mexico, obtained from the Federal Reserve Economic Data (FRED).

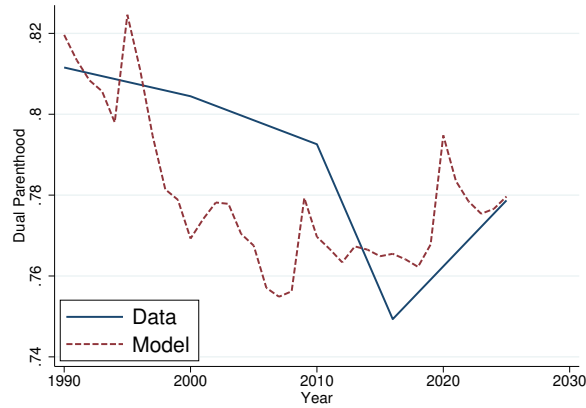
A direct comparison of levels is uninformative because the two objects measure different margins and live on different scales. The model’s $l_t^\varnothing$ is an intensive-margin time allocation that moves within a narrow band (approximately 0.78 to 0.90 over the sample), while FLFP is an extensive-margin participation rate measured in percentage points. To make the comparison meaningful, I normalize each series as a fraction of its total 1990–2025 change: both equal zero in 1990 and one in 2025. This rescaling preserves the *pace* of the transition—the fraction of the total gain accumulated by each date—while removing the level and unit differences.



(a) Productivity Inputs (A_t , B_t)



(b) Fertility



(c) Dual Parenthood

Figure 5. Model Fit: Mexico (1990–2025)

Notes: Solid lines denote data and dashed lines denote model values. Panel (a) shows the two productivity inputs: A_t (gender-neutral, solid) and B_t (relative wage gap, dashed). Fertility is children per woman. Dual parenthood measures children's co-residence with both parents; the model maps adult marriage and relative fertility into this child-based measure. Parameters appear in Table 6.

Figure 6 plots the two normalized series. Their correlation is 0.979. The model reproduces the rapid increase in female labor supply through approximately 2008 and the subsequent deceleration, matching both the timing and the curvature of the FLFP trajectory. The early acceleration reflects the sharp decline in the relative gender wage gap B_t , which raises the opportunity cost of children and draws women into market work. The post-2008 slowdown coincides with a moderating pace of wage convergence. Notably, the model is driven entirely by productivity, wage-gap, and childcare-division inputs calibrated to fertility and family structure—not to labor supply. Its ability to nonetheless generate a labor-supply path close to the independently observed FLFP series reassures us that the underlying mechanism operates through the channels the theory emphasizes.

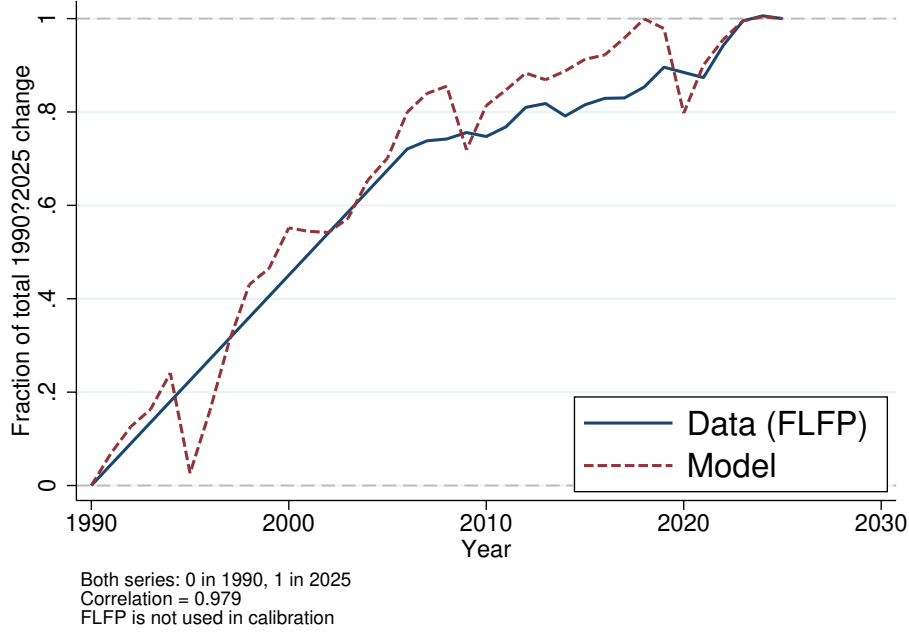


Figure 6. Out-of-Sample Validation: Female Labor Supply vs. FLFP

Notes: Both series are normalized as fractions of their respective total 1990–2025 changes (0 in 1990, 1 in 2025). The solid line is Mexico’s female labor force participation rate from FRED. The dashed line is the model’s predicted female labor supply, $l_t^{\circ} = 1 - \chi s_t n_t$. FLFP is not used in calibration. Correlation = 0.979.

5.3 Equilibrium Outcomes and Welfare

One advantage of the model is that it recovers intra-household transfers and agent-specific welfare, objects that are difficult to observe directly.

Within-marriage transfers

Figure 7a shows that the equilibrium transfer share α_t declines from approximately 4.4 percent to 3.2 percent over the sample. The fall is modest despite the sharp decline in the relative wage gap, because as the wage gap narrows the husband must transfer a larger share of a smaller relative income to sustain any given marital fertility through income effects. These two forces roughly offset in the share.

Consumption equivalents

I measure welfare using consumption equivalents benchmarked to the married male. For each agent type j , the consumption equivalent CE_j is the proportional increase in consumption that agent j would need, holding fertility and marital status fixed, to achieve the married male’s utility level. Formally, $CE_j = \mu_j - 1$, where μ_j solves $u(\mu_j c_j, n_j) = V^{\sigma, m}/\lambda$ when j is married (using the

systematic component) and the analogous expression for single agents. A positive consumption equivalent indicates that the agent is worse off than the married male and would require additional consumption to match his welfare; a negative value would indicate the reverse.¹⁵

Figure 7b reports the results. All three non-benchmark agent types have positive consumption equivalents throughout, indicating that all are worse off than married males. However, all three measures decline substantially over the sample, reflecting convergence toward married-male welfare.

Single women begin the sample with the largest gap: a consumption equivalent of 2.23, meaning they would need 223 percent more consumption to match the married male's welfare. By 2025, this falls to approximately 0.85. Married women start at 1.99 and decline to approximately 0.80. Both declines are driven primarily by the closing wage gap, which raises female wages and thus female consumption directly. Single men have the smallest gap (starting at 0.10 and declining to approximately 0.03) since they already earn the male wage and differ from married men only in lacking children and the marriage benefit.

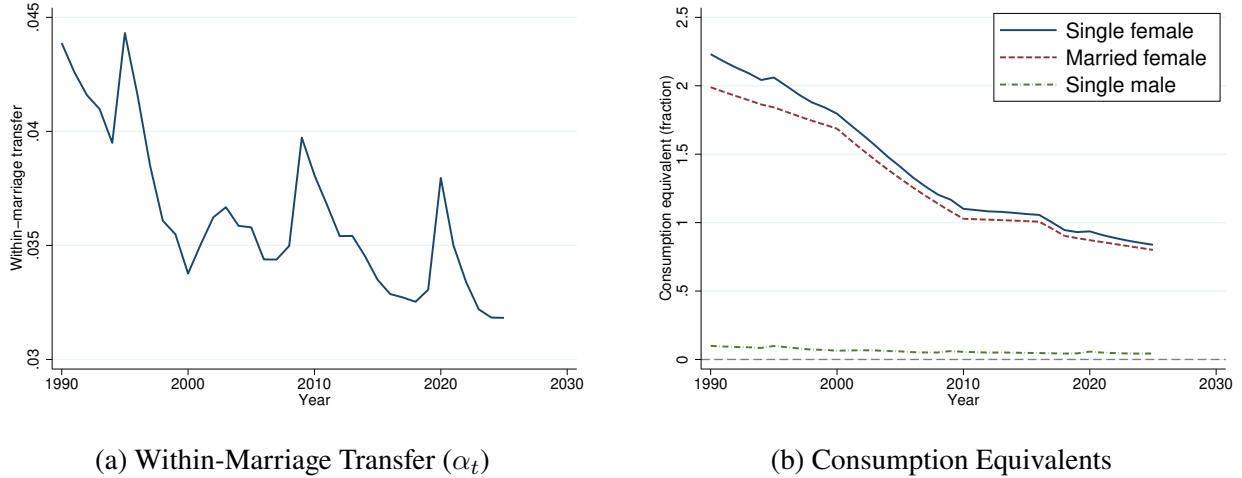


Figure 7. Equilibrium Outcomes and Welfare

Notes: Simulations use Table 6. Panel (a) shows the equilibrium transfer share α_t . Panel (b) reports consumption equivalents benchmarked to the married male. $CE > 0$ indicates the agent is worse off than the married male and needs additional consumption to match his welfare. The dashed horizontal line at zero marks the married-male benchmark.

Table 7 reports consumption equivalents at selected years. The married-female consumption equivalent declines by more than one full unit, from 1.99 in 1990 to 1.01 in 2015, indicating that the within-marriage welfare gap between spouses has roughly halved. Women's rising earning power, rather than larger transfers, drives this convergence.

¹⁵For the married female, the consumption equivalent simplifies to $CE_{\text{mf}}^{\sigma, m} = c_{\text{mf}}^{\sigma, m} / c_{\text{mm}}^{\sigma, m} - 1$, since married spouses share the same fertility. For single agents, the comparison accounts for different fertility levels using the CES aggregator.

Year	α	CE: Single female	CE: Married female	CE: Single male
1990	0.044	2.23	1.99	0.10
1995	0.044	2.06	1.84	0.10
2000	0.034	1.80	1.69	0.06
2005	0.036	1.41	1.32	0.06
2010	0.038	1.10	1.03	0.06
2015	0.033	1.06	1.01	0.05

Table 7. Equilibrium Transfer and Consumption Equivalents at Selected Years

Notes: α is the equilibrium transfer share. Consumption equivalents are benchmarked to the married male (CE = 0). Positive values indicate the agent needs additional consumption to match married-male welfare. Parameters from Table 6.

5.4 Childcare Redistribution Conditions

Using the calibrated equilibrium, I evaluate the three conditions under which a small redistribution of childcare from wives to husbands raises fertility, marriage, and gender income equality. The calibration assumes equal parental efficiency ($\eta = 1$), so that one hour of the father’s time produces the same effective care as one hour of the mother’s. Because the extended model has positive paternal childcare throughout ($\chi_t^{\sigma} > 0$), the general formulas in Appendix E.2 apply.

At $\eta = 1$, all three conditions are satisfied in every year from 1990 to 2025. Condition 1 ($w^{\varphi} > \alpha w^{\sigma}$) holds comfortably throughout: the equilibrium transfer share never exceeds 4.5 percent of male earnings, so the wife’s own wage far exceeds the transfer-weighted male wage and redistribution unambiguously lowers her effective childcare cost. Condition 2 (income convergence after fertility adjusts) is satisfied because the per-child redistribution of care more than compensates for the additional care generated by higher fertility. Condition 3 (the husband gains from the adjusted contract) holds because the narrowing wage gap keeps the earnings cost of paternal care moderate relative to the fertility gain that the husband values.

The joint satisfaction of all three conditions throughout the sample indicates that Mexico has been in the region where Proposition 5 predicts that childcare redistribution can simultaneously improve all three outcomes since at least 1990. This is consistent with the time-use evidence: the female childcare share has declined from 0.85 to 0.73 over the sample, and the conditions predict that further redistribution would continue to benefit the trinity.

Minimum paternal efficiency threshold

The conditions above are evaluated at $\eta = 1$, the value maintained throughout the calibration. A natural question is how robust the result is to lower paternal efficiency. To answer it, I conduct the following counterfactual: holding the calibrated preference and marriage parameters ($\beta, \rho, \lambda, \theta$) fixed at their $\eta = 1$ estimates (Table 6), I vary η on a fine grid in $(0, 1]$, set the father’s raw time to

χ_t^σ / η , re-solve the equilibrium at each year's observed wages and childcare division, and evaluate the three conditions of Proposition 5. The threshold $\bar{\eta}_t$ defined in (20) is the smallest η at which all three conditions still hold.¹⁶

Figure 8 plots the result. In 1990, $\bar{\eta} \approx 0.94$: fathers could be at most 6 percent less productive than mothers before the redistribution result fails. By the mid-2010s the threshold falls to approximately 0.77, meaning fathers could be 23 percent less efficient and redistribution would still improve all three outcomes. The decline reflects the narrowing wage gap, which reduces the earnings cost of paternal care. Because the husband's foregone earnings per effective unit of care equal w^σ / η , a lower wage gap relaxes Condition (19) and widens the range of admissible efficiencies. The binding constraint throughout is Condition 3 (the husband's participation gain); Condition 1 is never close to binding because the equilibrium transfer share is small.

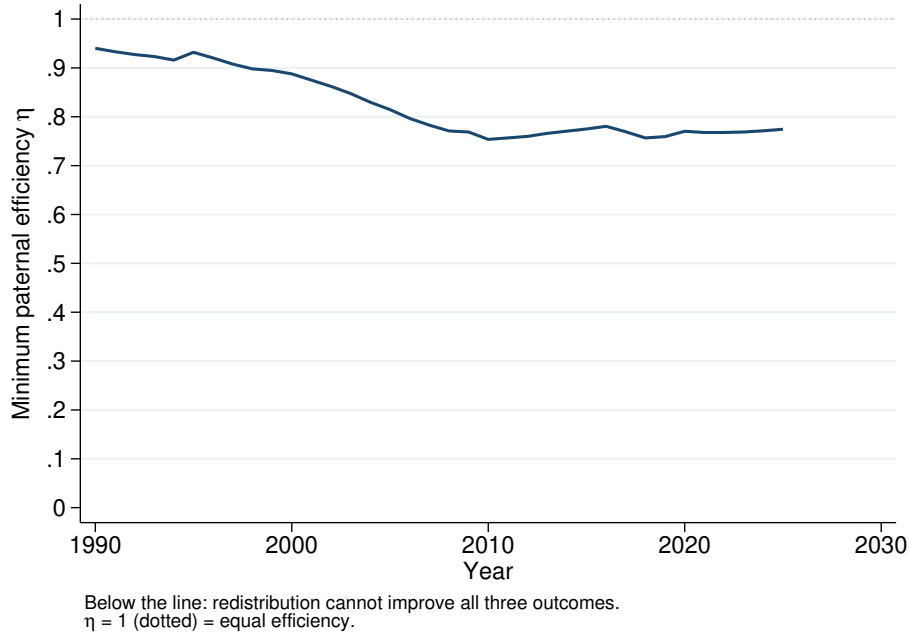


Figure 8. Minimum Paternal Efficiency Threshold $\bar{\eta}$ (Mexico, 1990–2025)

Notes: The solid line plots the minimum paternal care efficiency $\bar{\eta}$ such that all three conditions of Proposition 5 are satisfied. The exercise holds calibrated parameters at their $\eta = 1$ estimates (Table 6) and varies η at each year's equilibrium. Below the line, redistribution cannot improve all three outcomes. The dotted line at $\eta = 1$ marks equal efficiency. The decline in $\bar{\eta}$ reflects the narrowing gender wage gap, which reduces the earnings cost of paternal care and widens the scope for redistribution.

The declining threshold provides a quantitative perspective on the redistribution result. Even if fathers are modestly less efficient at childcare, the conditions for joint improvement hold throughout

¹⁶Because the preference and marriage parameters are held at their $\eta = 1$ values, the exercise isolates the effect of lower paternal efficiency on the redistribution conditions while keeping the rest of the economic environment fixed. A full re-calibration for each η would change the estimated parameters and is left for future work.

the sample. As economies develop and wage gaps narrow, the scope for redistribution widens. Appendix G confirms that these patterns—the satisfaction of all three conditions at $\eta = 1$ and the declining trajectory of $\bar{\eta}$ —are preserved under alternative values of the childcare time cost χ .

5.5 Shapley-Value Decomposition

I decompose changes in fertility, dual parenthood, the gender income gap, and consumption-equivalent welfare into contributions from three forces: gender-neutral productivity (A), the relative wage gap (B), and the gender division of childcare (χ).

Methodology

For each force, “on” means the variable takes its observed time-varying value; “off” means it is held fixed at its 1990 value. All $2^3 = 8$ combinations are solved. The Shapley formula assigns each force its average marginal contribution across all possible orderings of the three forces. The decomposition is exact: the three Shapley contributions sum to the total change from the all-off baseline in every year. Unlike a sequential decomposition, the Shapley value is invariant to the ordering of forces.¹⁷

Results

Figure 9 presents the decomposition for fertility, dual parenthood, the gender income gap, and the married-female consumption equivalent. Table 8 reports the Shapley shares at the end of the sample.

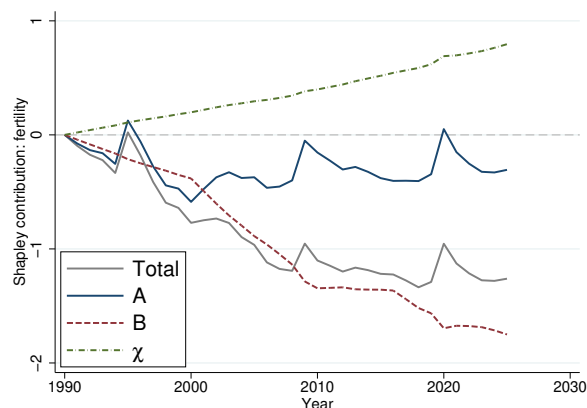
Outcome	A (%)	B (%)	χ (%)
Fertility	24.3	138.8	−63.1
Dual parenthood	55.0	212.5	−167.4
Gender income gap	3.3	94.5	2.2
CE: Single female	2.4	97.7	−0.2
CE: Married female	0.5	107.2	−7.6
CE: Single male	21.8	82.6	−4.4

Table 8. Shapley Shares at 2025 (% of Total Change from 1990 Baseline)

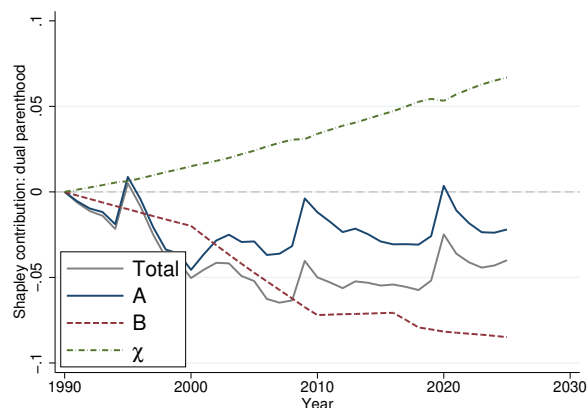
Notes: Each row reports the Shapley share of the three forces as a percentage of the total change from the baseline (all inputs at 1990 values). Shares sum to 100 percent in each row. A negative share indicates that the force moves the outcome in the opposite direction from the total change.

Three findings stand out.

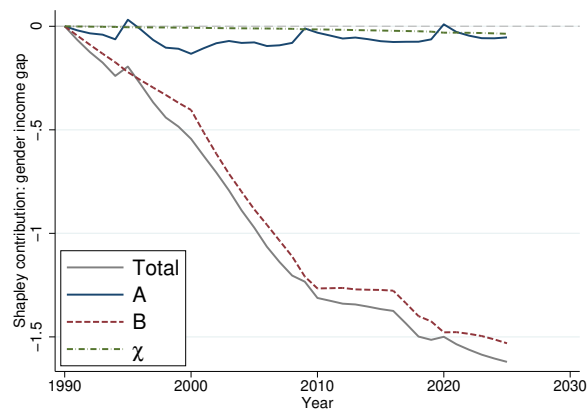
¹⁷For three forces with scenario values $V_0, \dots, V_7$, the Shapley value of force A is $\phi_A = \frac{1}{3}(V_4 - V_0) + \frac{1}{6}(V_6 - V_2) + \frac{1}{6}(V_5 - V_1) + \frac{1}{3}(V_7 - V_3)$, with analogous expressions for B and χ . The exactness property $\phi_A + \phi_B + \phi_\chi = V_7 - V_0$ holds by construction.



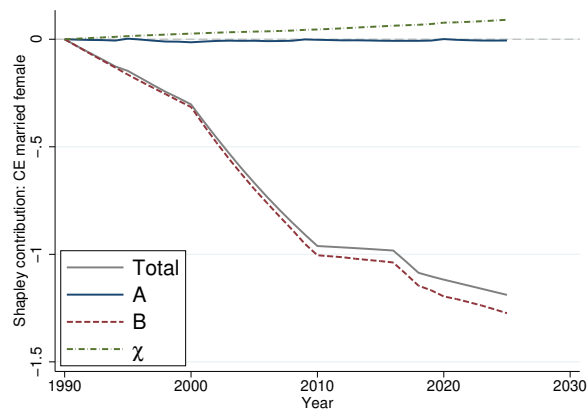
(a) Fertility



(b) Dual Parenthood



(c) Gender Income Gap



(d) CE: Married Female

Figure 9. Shapley-Value Decomposition of Mexico's Transition (1990–2025)

Notes: Each panel plots the total change from the all-off baseline (gray solid) and the three Shapley contributions: *A* (gender-neutral productivity, navy solid), *B* (relative wage gap, maroon dashed), and χ (childcare division, green dash-dot). The dashed horizontal line at zero marks the no-change reference. The decomposition is exact in every year. Appendix Figure A.4 reports analogous decompositions for single-female and single-male consumption equivalents.

The declining relative wage gap is the dominant driver. For every outcome, *B* (the narrowing gap between male and female wages) accounts for the largest share, and in most cases the overwhelming majority, of the total change. It explains 139 percent of the fertility decline (more than the total because the childcare-division force partially offsets it), 95 percent of income convergence, and 97–107 percent of the decline in female consumption equivalents. Higher female wages raise the opportunity cost of children, reduce the income-sharing attraction of marriage, and directly narrow the income gap.

More equal childcare partially buffers fertility and marriage. The Shapley contribution of χ is negative for fertility (−63 percent) and dual parenthood (−167 percent), meaning that the shift toward more equal childcare *offsets* a substantial fraction of the declines driven by *B*. This

is precisely the mechanism identified in Proposition 5: redistributing childcare from mothers to fathers lowers the wife’s effective childcare cost, raises marital fertility, and supports marriage, partially counteracting the fertility- and marriage-depressing effects of wage convergence.

Gender-neutral productivity plays a supporting role. Rising A contributes modestly to the fertility and dual-parenthood declines (24 and 55 percent, respectively) by raising the opportunity cost of children, but it accounts for very little of the income-gap or welfare convergence (3 percent and below 1 percent for females, 22 percent for single males). The income-gap result reflects that a proportional rise in productivity raises both male and female wages similarly, leaving relative earnings nearly unchanged.

The decomposition reveals that the evolving division of childcare is quantitatively important for understanding fertility and family structure, even though it contributes little to income convergence. The childcare channel operates primarily through the household margin (lowering the effective price of children within marriage) while the income-gap margin is dominated by the direct wage effect of B .

6 Conclusion

This paper establishes a three-way trade-off among fertility, dual parenthood, and gender income equality. Data from 115 countries show that societies achieving any two almost invariably fall short on the third. The shortfall relative to statistical independence is remarkably similar in OECD and non-OECD countries, yet an exact pairwise decomposition traces it to different sources: the fertility–dual parenthood tension dominates in the OECD, while the fertility–gender equality tension dominates elsewhere. A tractable marriage-market model explains the contrast. Because marriage bundles an income transfer with shared fertility, the equilibrium contract links all three outcomes, and the gender wage gap Γ^w determines which pairwise tension binds. At high wage gaps, fertility strongly amplifies income inequality through forgone female earnings, tying the binding constraint to gender equality. As the gap narrows, that amplification weakens and the fertility–marriage tension emerges instead.

Three conventional instruments—pro-marriage benefits, higher female wages, and lower child-care costs—each improve two objectives at the expense of the third. Coordinating a wage increase with a marriage-benefit adjustment can in principle achieve all three improvements at any interior equilibrium, but the required calibration is exacting. The paper’s central applied contribution is a simpler, single-instrument alternative: redistributing childcare from wives to husbands. The reform succeeds when it lowers the wife’s effective cost of children, preserves a gain for the husband after transfers adjust, and leaves relative earnings more equal once fertility responds. These bilateral conditions explain why similar paternity-leave reforms can produce different demographic

responses across settings: both spouses must gain from the modified marital contract.

The analysis allows fathers to be less efficient than mothers at providing care. A minimum efficiency threshold $\bar{\eta}$ characterizes how much room for redistribution exists: when $\eta \geq \bar{\eta}$ the reform improves all three outcomes; when $\eta < \bar{\eta}$ at least one condition fails. In the Mexico calibration, a counterfactual exercise that varies η while holding other parameters at their $\eta = 1$ estimates finds that $\bar{\eta}$ declines from 0.94 in 1990 to approximately 0.77 by the mid-2010s. The narrowing wage gap reduces the earnings cost of paternal care, widening the set of efficiencies under which redistribution works. This declining threshold suggests that the scope for redistribution grows as economies develop and gender wage gaps converge.

The quantitative exercise reinforces the theory. Calibrated to Mexico from 1990 to 2025, the extended model closely tracks fertility and family structure, and its implied female labor supply reproduces the trajectory of Mexico's FLFP rate as a non-targeted moment. A Shapley-value decomposition attributes the bulk of Mexico's demographic transitions to the declining relative wage gap, while the increasingly equal division of childcare partially offsets the fertility and dual-parenthood declines—directly illustrating the redistribution mechanism. All three conditions for childcare redistribution to improve the trinity are satisfied throughout the sample.

Two broader lessons emerge. First, sustained progress along all three dimensions is possible only when both spouses gain from the reform. Policies that quietly rely on one partner to absorb the cost of the other's gains tend to fail at precisely the participation margin they need to move. Second, the welfare effects of family policy cannot be assessed in partial equilibrium. The within-family transfers that clear the marriage market respond endogenously to policy, and those responses can reverse the sign of the welfare effect that a partial-equilibrium analysis would predict.

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Appendix

A Additional Empirical Results

A.1 Binarization Cutoffs and Sample Diagnostics

Table A.1 reports the exact binarization cutoffs for each sample, alongside summary statistics for the three outcome variables. A country-year is classified as “high” on a dimension if and only if its value strictly exceeds the sample-specific cutoff.

Sample	Countries	Obs	TFR		Both parents (%)		Female income share	
			Median	SD	Median	SD	Median	SD
Full	115	1,764	2.00	1.50	80.1	12.5	0.341	0.079
OECD	39	879	1.56	0.33	83.7	5.6	0.374	0.058
Non-OECD	76	885	3.24	1.52	70.5	14.3	0.294	0.080

Table A.1. Binarization Cutoffs and Sample Statistics

Notes: Cutoffs are the sample-specific medians in original units. TFR is children per woman. “Both parents” is the share of children living with both parents (percentage points). Female income share is the female share of national labor income (proportion). SD denotes the standard deviation within each sample. A country-year is classified as “high” on a dimension if its value strictly exceeds the corresponding cutoff.

Table A.2 reports pairwise correlations in original units for each sample. All three pairs are negatively correlated in all samples. The strongest correlation in the full sample is between fertility and the female income share (-0.52). In the OECD, the fertility–gender equality and dual parenthood–gender equality correlations are both moderately negative (-0.42 and -0.31). In non-OECD countries, the dual parenthood–gender equality correlation is the strongest (-0.40), while fertility and dual parenthood are only weakly negatively correlated (-0.12).

Sample	TFR–BPP	TFR–FIS	BPP–FIS
Full	-0.41	-0.52	-0.05
OECD	-0.26	-0.42	-0.31
Non-OECD	-0.12	-0.33	-0.40

Table A.2. Pairwise Correlations in Original Units

Notes: TFR is the total fertility rate. BPP is the share of children living with both parents. FIS is the female income share.

A.2 Bootstrap Inference

All bootstrap exercises use 1,000 replications clustered by country. Table A.3 reports detailed bootstrap results for the trinity shortfall. Table A.4 reports bootstrap results for the pairwise contribution decomposition.

Statistic	Full		OECD		Non-OECD	
	Estimate	Boot SE	Estimate	Boot SE	Estimate	Boot SE
Trinity share	0.028	0.014	0.060	0.020	0.061	0.023
Shortfall (pp)	−9.65	1.40	−6.34	2.04	−6.38	2.30
$I(F; M)$ (bits)	0.121	0.054	0.076	0.054	0.014	0.025
$I(F; G)$ (bits)	0.239	0.066	0.000	0.017	0.060	0.044
$I(M; G)$ (bits)	0.015	0.018	0.038	0.037	0.027	0.033
Total correlation (bits)	0.372	0.077	0.119	0.071	0.128	0.056
Interaction information (bits)	0.003	0.029	−0.005	0.023	−0.028	0.026

Table A.3. Bootstrap Inference for Key Statistics

Notes: Bootstrap uses 1,000 replications clustered by country. Shortfall is $p_{111} - p_{FPM}p_G$ in percentage points. $I(\cdot; \cdot)$ denotes pairwise mutual information. Total correlation is $H(F) + H(M) + H(G) - H(F, M, G)$. Interaction information is $I(F; M) - I(F; M | G)$; negative values indicate synergy, positive values indicate redundancy. Interaction information estimates are not statistically significant at the 5 percent level in any sample.

Component	OECD				Non-OECD			
	Point (pp)	Boot SE	95% CI	p	Point (pp)	Boot SE	95% CI	p
F-M tension	−4.02	1.35	[−6.63, −1.50]	0.003	−1.71	1.49	[−4.29, 1.52]	0.253
F-G tension	+0.07	1.56	[−3.64, 2.39]	0.964	−3.57	1.28	[−5.99, −0.97]	0.005
M-G tension	−2.86	1.42	[−5.34, 0.11]	0.044	−2.42	1.49	[−4.96, 0.72]	0.105
Three-way	+0.47	1.24	[−2.07, 2.81]	0.706	+1.31	1.26	[−0.96, 3.90]	0.300

Table A.4. Bootstrap Inference for the Pairwise Contribution Decomposition

Notes: Point estimates and bootstrap standard errors are in percentage points. Confidence intervals are percentile-based. The p -value is from a two-sided normal approximation using the point estimate and bootstrap standard error. The F-M tension is significantly negative in the OECD; the F-G tension is significantly negative in the non-OECD. The F-G tension in the OECD is indistinguishable from zero.

The interaction information estimates are negative in the OECD (−0.005 bits) and non-OECD (−0.028 bits) subsamples, suggesting synergy, but neither is statistically significant at the 5 percent level. Rather than emphasizing the sign of interaction information, the paper focuses on the statistically significant trinity shortfall and its decomposition into pairwise tensions, which provides a more directly interpretable characterization of the trade-off.

A.3 Robustness: Gender Wage Gap

The female income share used in the main analysis reflects both wages and labor-force participation. Because higher fertility reduces mothers' market work, the female income share can fall mechanically even if wages are unchanged. This raises the concern that part of the measured fertility–gender equality tension operates through participation rather than through the wage structure itself.

To address this concern, I replace the WID female income share with the OECD unadjusted gender wage gap, defined as the difference between male and female median full-time earnings as a percentage of male median earnings. This measure conditions on employment and therefore holds the participation margin fixed. I negate the wage gap so that higher values indicate greater equality, paralleling the female income share. The robustness sample comprises OECD countries for which wage-gap, dual-parenthood, and TFR data are jointly available. All medians and binary indicators are re-estimated within this sample.

Table A.5 reports the trinity shortfall. The observed joint-attainment share is 8.0 percent, compared with 12.4 percent under independence, yielding a shortfall of 4.43 percentage points. The shortfall is smaller than in the main OECD sample (6.34 percentage points), consistent with the removal of the participation channel that mechanically links fertility to measured gender inequality. The observed-to-independence ratio is 0.64, still well below unity.

Sample	Gender equality measure	Observed (%)	Independence (%)	Shortfall (pp)
OECD (main)	Female income share (WID)	6.03	12.37	−6.34
OECD (robustness)	Neg. gender wage gap (OECD)	8.0	12.4	−4.43

Table A.5. Trinity Shortfall: Main vs. Robustness Gender Equality Measure

Notes: The main OECD sample uses the WID female income share, which incorporates differences in labor-force participation. The robustness sample uses the OECD unadjusted gender wage gap among full-time employed workers (negated so that higher values indicate greater equality), which conditions on employment and removes the participation margin. Both samples use TFR for fertility and children's co-residence with both parents for dual parenthood. All medians are re-estimated within each sample.

Figure A.1 presents the Venn diagram for the robustness sample. The two largest categories are “F only” (18.4 percent) and “M+G” (17.7 percent), preserving the pattern from the main OECD analysis: countries with high fertility tend to lack dual parenthood and gender equality, while countries with both tend to have low fertility. The “F+G” category (16.5 percent) is notably larger than in the main OECD sample (19.0 percent for the same category), reflecting the weaker negative association between fertility and gender equality once the participation channel is removed.

Table A.6 reports the pairwise contribution decomposition and Figure A.2 compares it with the main OECD analysis. The fertility–dual parenthood tension now accounts for virtually the entire shortfall, while the fertility–gender equality and dual parenthood–gender equality tensions are both small. This pattern is consistent with the interpretation that, once the participation channel is

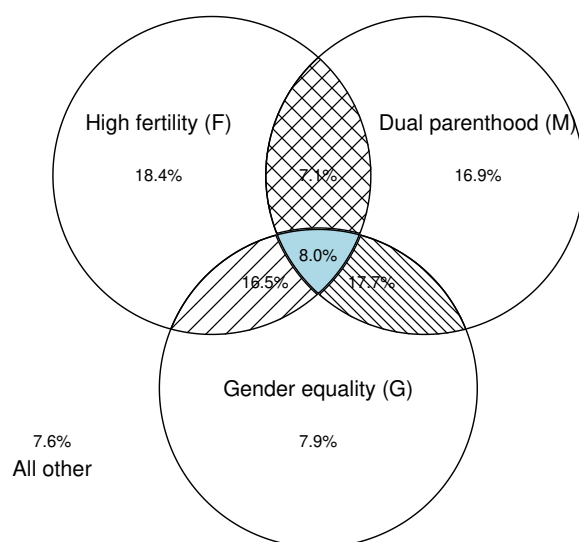


Figure A.1. Three-Way Trade-Off: OECD Robustness Sample (Gender Wage Gap)

Notes: Venn diagram for the OECD robustness sample using the negated gender wage gap among employed workers as the gender equality measure. Each dimension is split at the sample-specific median. Under statistical independence, the central overlap (F+M+G) would contain approximately 12.5 percent of observations.

removed, fertility and gender equality largely decouple in the OECD. The residual tension between fertility and dual parenthood, which operates through the marriage market and the composition of families rather than through female labor supply, remains the binding constraint.

Component	OECD (income share)		OECD (wage gap)	
	Contribution (pp)	% of shortfall	Contribution (pp)	% of shortfall
F-M tension	-4.02	63.4%	-4.43	≈100%
F-G tension	+0.07	-1.1%	-0.27	≈6%
M-G tension	-2.86	45.1%	+0.22	≈-5%
Three-way residual	+0.47	-7.4%	+0.05	≈-1%
Total shortfall	-6.34	100%	-4.43	100%

Table A.6. Pairwise Contribution Decomposition: Main vs. Robustness Measure

Notes: The left panel reproduces the main OECD decomposition from Table 4. The right panel uses the negated OECD gender wage gap among employed workers. Approximate percentages for the robustness sample are read from the decomposition output; see Figure A.2 for the graphical comparison.

Two features of the comparison merit emphasis. First, the M–G tension, which accounts for 45.1 percent of the shortfall in the main analysis, essentially disappears with the wage-gap measure. This is consistent with the interpretation that much of the measured tension between dual parenthood and gender equality in the main analysis operates through participation: marriage raises fertility,

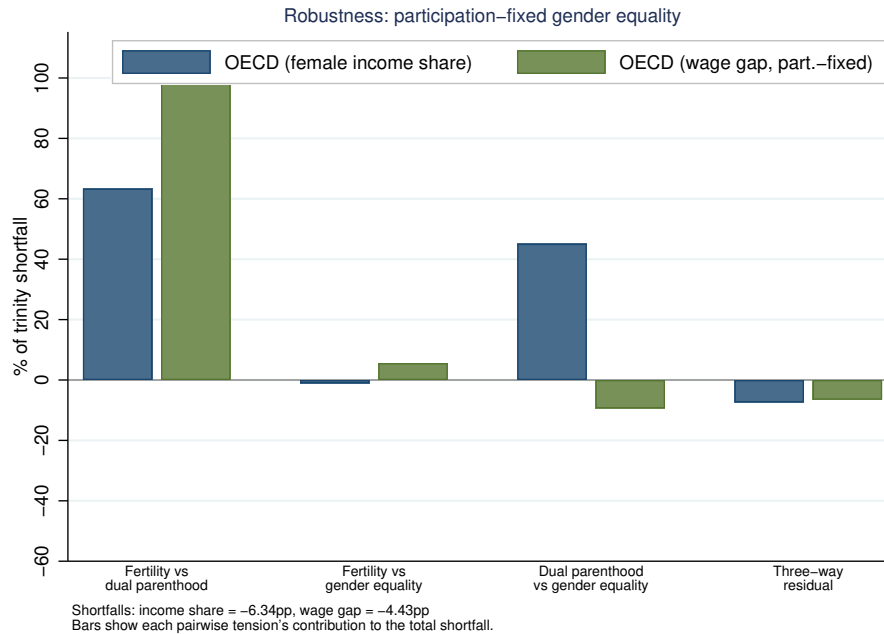


Figure A.2. Decomposition Comparison: Female Income Share vs. Gender Wage Gap

Notes: Bars show each pairwise tension's contribution to the total trinity shortfall, expressed as a percentage. Navy bars use the WID female income share (main analysis); green bars use the negated OECD gender wage gap among employed workers (robustness). The wage-gap measure conditions on employment, removing the mechanical channel through which fertility reduces measured gender equality via lower female labor-force participation.

which reduces female labor supply and lowers the income share. Conditioning on employment removes this channel. Second, the F–G tension remains negligible under both measures, confirming that fertility and gender equality among workers are approximately orthogonal in the OECD. The dominant constraint on the trinity in OECD countries is the fertility–dual parenthood tension, regardless of which gender equality measure is used.

B Additional Figures

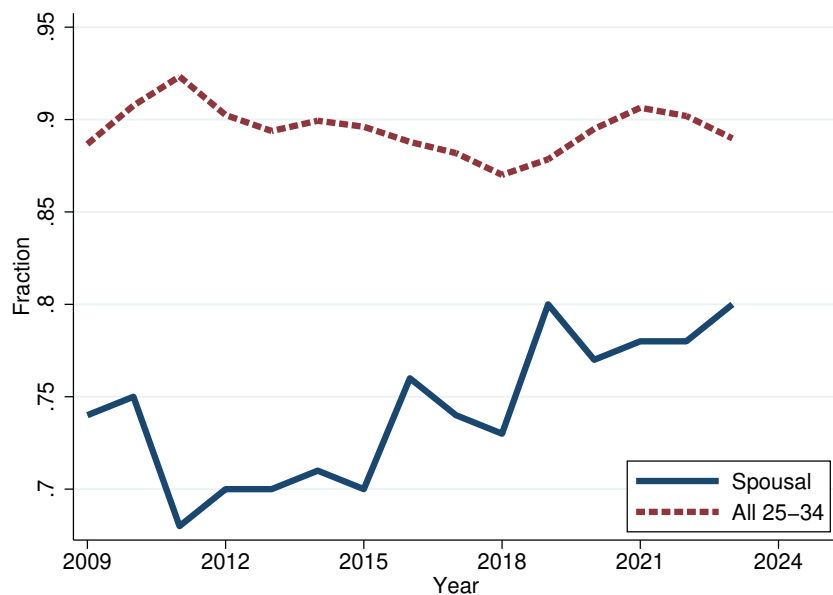
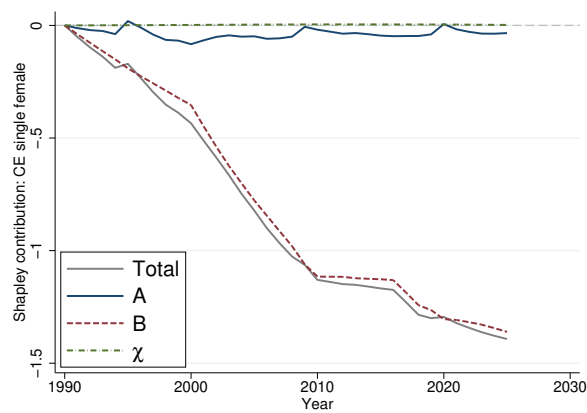
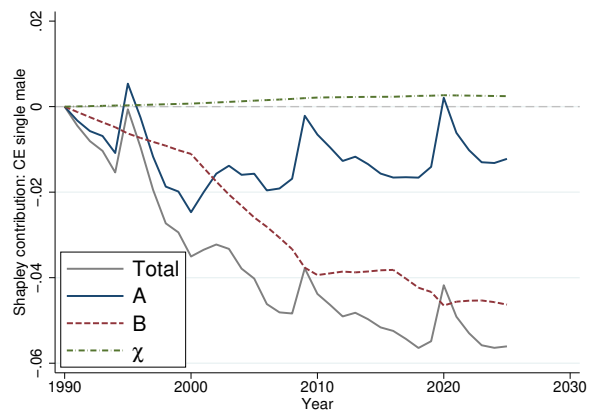


Figure A.3. Gender Earnings Gaps in the U.S.

Notes: The spousal earnings gap uses ACS women aged 18–39, not enrolled in school, childless, and in a first marriage within the past year to a husband also not enrolled in school. The overall gap for full-time wage and salary workers aged 25–34 is from the BLS Annual Highlights of Women’s Earnings.



(a) Shapley Decomposition: CE Single Female



(b) Shapley Decomposition: CE Single Male

Figure A.4. Additional Results on Consumption Equivalents

Notes: Each panel plots the total change from the all-off baseline (gray solid) and the three Shapley contributions: A (gender-neutral productivity, navy solid), B (relative wage gap, maroon dashed), and χ (childcare division, green dash-dot). The dashed horizontal line at zero marks the no-change reference. See Table 8 for the Shapley shares at 2025.

C Identification of Calibrated Parameters

The calibration estimates four parameters $(\beta, \rho, \lambda, \theta)$ by matching four moments: mean fertility, the fertility change over the sample, mean dual parenthood, and the dual-parenthood change. This appendix examines local identification by computing the elasticity Jacobian of the moment vector with respect to the parameter vector and its inverse, both evaluated at the calibrated values in Table 6.

Forward Jacobian

Define the elasticity Jacobian $\mathcal{J}$ with entries $\mathcal{J}_{ij} = (\partial m_i / \partial p_j)(p_j / m_i)$, where m_i is the i -th moment and p_j the j -th parameter. The matrix is computed by perturbing each parameter by one percent from its calibrated value and recording the resulting percentage change in each moment. With rows ordered as (mean fertility, fertility change, mean dual parenthood, dual-parenthood change) and columns as $(\beta, \rho, \lambda, \theta)$:

$$\mathcal{J} = \begin{pmatrix} 2.689 & 1.131 & 1.209 & 0.073 \\ 2.161 & 2.989 & -1.104 & 0.057 \\ 0.676 & 0.108 & 2.539 & 0.157 \\ 4.355 & 1.698 & -7.113 & 2.001 \end{pmatrix}. \quad (\text{C.1})$$

The diagonal entries confirm that each parameter has its strongest absolute effect on the intended target: β on mean fertility (2.689), ρ on the fertility change (2.989), λ on mean dual parenthood (2.539), and θ on the dual-parenthood change (2.001). Several off-diagonal entries are nonnegligible, as expected in a model where all outcomes are jointly determined. In particular, λ exerts a large influence on the dual-parenthood change (-7.113), reflecting that the marriage-benefit parameter shifts the *level* of dual parenthood, which in turn moves the endpoint difference. The ratio of the diagonal entry to the largest off-diagonal entry in the same row provides a simple measure of identification strength: this ratio is 2.38 for β , 1.38 for ρ , 3.75 for λ , and only 0.28 for θ . The last value indicates that θ 's intended target moment is more sensitive to λ than to θ itself.

Inverse Jacobian

The inverse $\mathcal{J}^{-1}$ reverses the mapping: its (i, j) entry gives the percentage adjustment in parameter i required to restore the calibration when moment j shifts by one percent. The rows of $\mathcal{J}^{-1}$ therefore

reveal which moments carry the most information about each parameter:

$$\mathcal{J}^{-1} = \begin{pmatrix} 0.631 & -0.232 & -0.367 & 0.012 \\ -0.459 & 0.510 & 0.366 & -0.027 \\ -0.072 & 0.029 & 0.366 & -0.027 \\ -1.239 & 0.177 & 1.787 & 0.400 \end{pmatrix}. \quad (\text{C.2})$$

The first row shows that β is primarily informed by mean fertility (0.631), with a moderate negative entry on mean dual parenthood (-0.367) reflecting the indirect channel through which β affects family composition, and a negligible entry on the dual-parenthood change (0.012). The second row shows that ρ draws from several moments: the fertility change has the largest entry (0.510), consistent with the intended identification, but mean fertility (-0.459) and mean dual parenthood (0.366) carry nearly as much weight, reflecting ρ 's role in governing the curvature of the consumption–fertility tradeoff across both level and trend. The third row provides the cleanest case: λ is almost entirely pinned down by mean dual parenthood (0.366), with all other entries at most 0.072 in absolute value. This sharp identification is consistent with λ 's high diagonal-dominance ratio in the forward Jacobian.

The fourth row is the most informative. The dual-parenthood change, the moment intended to identify θ , enters with a weight of only 0.400, while mean dual parenthood (1.787) and mean fertility (-1.239) carry substantially larger absolute weights. Because λ has a stronger effect on the dual-parenthood change than θ does, the inverse assigns θ most of its identifying variation from moments that primarily discipline other parameters. Small measurement errors in mean dual parenthood or mean fertility therefore propagate into disproportionately large revisions of θ , implying that bootstrap or resampling exercises should produce wider confidence intervals for θ than for the other three parameters. This fragility does not compromise the model's ability to match the targeted moments in estimation, but it suggests that the dual-parenthood change provides limited independent leverage for disciplining participation responsiveness.

D Proofs for the Baseline Model

D.1 Marriage Probabilities and the Husband's Participation Relation

Let the independent shocks have distribution $F(\varepsilon) = \exp(-\varepsilon^{-\theta})$ for $\varepsilon > 0$. An individual marries if $\varepsilon^m V^{g,m} \geq \varepsilon^s V^{g,s}$. Integrating over $\varepsilon^s = x$ gives

$$\begin{aligned} \Pr(\varepsilon^m V^{g,m} < \varepsilon^s V^{g,s}) &= \int_0^\infty F\left(x \frac{V^{g,s}}{V^{g,m}}\right) dF(x) \\ &= \frac{(V^{g,s})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta}. \end{aligned}$$

Taking the complement yields (3). Strict monotonicity in the utility ratio gives (4).

Single men's value is $(1 - \beta)^{1/r} w^{\sigma^\circ}$. Therefore

$$\frac{V^{\sigma^\circ, m}}{V^{\sigma^\circ, s}} = \lambda \left[(1 - \alpha)^r + \frac{\beta}{1 - \beta} \left(\frac{n^m}{w^{\sigma^\circ}} \right)^r \right]^{1/r}. \quad (\text{D.1})$$

Setting this equal to $(\mathcal{M}/(1 - \mathcal{M}))^{1/\theta}$ gives the conditional participation schedule

$$\alpha = 1 - \left[\left(\frac{1}{\lambda} \left(\frac{\mathcal{M}}{1 - \mathcal{M}} \right)^{1/\theta} \right)^r - \frac{\beta}{1 - \beta} \left(\frac{n^m}{w^{\sigma^\circ}} \right)^r \right]^{1/r},$$

where its bracket is positive. At zero fertility this gives $\alpha = 1 - \lambda^{-1}(\mathcal{M}/(1 - \mathcal{M}))^{1/\theta}$, which equals zero only when $\mathcal{M}/(1 - \mathcal{M}) = \lambda^\theta$. This schedule describes the husband's willingness to participate conditional on fertility; the wife's household problem determines fertility.

D.2 Household Choices and Proof of Lemma 2

Proof. For a woman with budget $c + w^\circ \chi n = I$, the interior first-order condition is

$$\frac{u_n}{u_c} = \frac{\beta}{1 - \beta} \left(\frac{c}{n} \right)^{1/\rho} = w^\circ \chi.$$

It follows that

$$\frac{c}{n} = \left(\frac{(1 - \beta)w^\circ \chi}{\beta} \right)^\rho,$$

and hence

$$n \left[\left(\frac{(1 - \beta)w^\circ \chi}{\beta} \right)^\rho + w^\circ \chi \right] = I.$$

Setting $I = w^{\varnothing}$ in singlehood and $I = w^{\varnothing} + \alpha w^{\sigma}$ in marriage gives (7). Both consumption and fertility scale by $1 + \alpha\Gamma^w$ across states. Homogeneity therefore implies

$$\frac{V_{\varnothing,m}^{\varnothing}}{V_{\varnothing,s}^{\varnothing}} = \lambda(1 + \alpha\Gamma^w).$$

At the equilibrium of Lemma 1, $\alpha > 0$, so marital fertility exceeds single fertility. Substitution into (3) proves the marriage-rate expression in (10). $\square$

D.3 Proof of Lemma 1

Proof. Combine (D.1), the wife's utility ratio, and (4). Canceling λ and raising both sides to power r gives

$$(1 - \alpha)^r + \frac{\beta}{1 - \beta} \left(\frac{n^m}{w^{\sigma}} \right)^r = (1 + \alpha\Gamma^w)^r.$$

Substitute $n^m = (1 + \alpha\Gamma^w)/C$ to obtain

$$(1 - \alpha)^r = K(1 + \alpha\Gamma^w)^r.$$

Assumption 1 implies $0 < K < 1$. On $-1/\Gamma^w < \alpha < 1$, taking positive roots is valid:

$$1 - \alpha = K^{1/r}(1 + \alpha\Gamma^w).$$

This linear equation has the unique solution in (9), and substitution gives marital fertility.

The time restriction in (8) guarantees positive married female labor supply. Single female labor supply is positive because $C > \chi$, male consumption is positive because $\alpha < 1$, and female consumption is positive by the first-order condition. CES preferences are strictly quasiconcave on the positive domain, so the feasible interior solutions are unique household optima.

Every interior equilibrium satisfies the preceding equivalent equations. The constructed solution is therefore the unique interior equilibrium. The parameters λ and θ affect participation, while the transfer and fertility formulas depend on the remaining primitives. $\square$

D.4 Aggregate Formulas and the Empirical Family-Structure Mapping

For compactness define

$$g \equiv K^{1/r}, \quad q \equiv 1 + \alpha^*\Gamma^w = \frac{1 + \Gamma^w}{1 + g\Gamma^w} > 1.$$

Then

$$n^s = \frac{1}{C}, \quad n^m = \frac{q}{C}, \quad \mathcal{M} = \frac{(\lambda q)^\theta}{1 + (\lambda q)^\theta}, \quad n = \frac{1 + \mathcal{M}(q - 1)}{C}. \quad (\text{D.2})$$

The child-weighted empirical counterpart of marriage is

$$\frac{\mathcal{M}n^m}{n} = \frac{\mathcal{M}q}{1 + \mathcal{M}(q - 1)}.$$

This expression uses the existing outcome notation. Its partial derivatives are

$$\frac{\partial}{\partial \mathcal{M}} \frac{\mathcal{M}q}{1 + \mathcal{M}(q - 1)} = \frac{q}{[1 + \mathcal{M}(q - 1)]^2} > 0, \quad \frac{\partial}{\partial q} \frac{\mathcal{M}q}{1 + \mathcal{M}(q - 1)} = \frac{\mathcal{M}(1 - \mathcal{M})}{[1 + \mathcal{M}(q - 1)]^2} > 0. \quad (\text{D.3})$$

Thus marriage and its child-based counterpart move together whenever $\mathcal{M}$ and q move in the same direction, or one is unchanged and the other moves strictly. This applies to the individual-policy results. The relationship concerns their comparative statics; their levels reflect different population weights.

D.5 Proof of Lemma 3

Proof. Part (i) is immediate from (5). Solving $\Gamma^w/(1 - \chi n) \leq \bar{\Gamma}$ for n gives $\bar{n}(\Gamma^w) = (1/\chi)(1 - \Gamma^w/\bar{\Gamma})$, which is linear and strictly decreasing in Γ^w .

For part (ii), recall from Appendix D.4 that $q - 1 = (1 - g)\Gamma^w/(1 + g\Gamma^w)$ with $g = K^{1/r}$. The direct partial derivative is

$$\frac{\partial(q - 1)}{\partial \Gamma^w} = \frac{1 - g}{(1 + g\Gamma^w)^2} > 0.$$

Raising Γ^w by lowering $w^\varnothing$ at fixed w^σ reduces C (since $\rho > 1$), which reduces K and hence g . Because

$$\frac{\partial(q - 1)}{\partial g} = -\frac{\Gamma^w(1 + \Gamma^w)}{(1 + g\Gamma^w)^2} < 0,$$

the indirect channel also raises $q - 1$. Marriage $\mathcal{M} = (\lambda q)^\theta/[1 + (\lambda q)^\theta]$ is increasing in q , and $\varepsilon_{n,\mathcal{M}} = \mathcal{M}(q - 1)/[1 + \mathcal{M}(q - 1)]$ is increasing in both $\mathcal{M}$ and $q - 1$. $\square$

D.6 Proof of Proposition 1

Proof. The objects C , q , and household fertilities are independent of λ . Therefore

$$\mathcal{M}_\lambda = \frac{\theta \mathcal{M}(1 - \mathcal{M})}{\lambda} > 0, \quad n_\lambda = (n^m - n^s)\mathcal{M}_\lambda > 0.$$

With wages and χ fixed, $\Gamma^y = \Gamma^w / (1 - \chi n)$ rises. Since q is fixed, (D.3) also shows that the empirical family-structure counterpart rises. $\square$

D.7 Proof of Proposition 2

Proof. At fixed w^σ , an increase in w^φ raises C , K , and g , and lowers Γ^w . Since

$$q_{\Gamma^w} = \frac{1 - g}{(1 + g\Gamma^w)^2} > 0, \quad q_g = -\frac{(1 + \Gamma^w)\Gamma^w}{(1 + g\Gamma^w)^2} < 0,$$

both changes reduce q , so marriage falls. Both $n^s = 1/C$ and $n^m = q/C$ fall. Aggregate fertility satisfies

$$n_{w^\varphi} = \mathcal{M}(n^m)_{w^\varphi} + (1 - \mathcal{M})(n^s)_{w^\varphi} + (n^m - n^s)\mathcal{M}_{w^\varphi} < 0.$$

Female wages and female labor supply both rise, so Γ^y falls. The child-based family-structure measure falls because $\mathcal{M}$ and q both fall. These conclusions hold for either sign of α_{w^φ} . $\square$

D.8 Proof of Proposition 3

Proof. Reducing χ lowers C , K , and g at fixed Γ^w , so q and marriage rise. Household fertilities rise, and the shift toward marriage reinforces aggregate fertility.

To determine the income response, write

$$\chi n = \frac{\chi}{C}[1 + \mathcal{M}(q - 1)], \quad \frac{\chi}{C} = \frac{1}{1 + \left(\frac{1-\beta}{\beta}\right)^\rho (w^\varphi \chi)^{\rho-1}}.$$

Since $\rho > 1$, the first factor rises as χ falls. The second also rises because $\mathcal{M}$ and q increase and $q > 1$. Hence female labor supply falls and the income gap widens. The empirical family-structure counterpart rises by (D.3). $\square$

D.9 Conditional Feasibility and Proof of Proposition 4

Derivation of (11). Equation (10) gives

$$\frac{1}{\lambda} \left(\frac{\mathcal{M}}{1 - \mathcal{M}} \right)^{1/\theta} = 1 + \alpha^* \Gamma^w.$$

Since $\alpha^* < 1$,

$$1 + \alpha^* \Gamma^w < 1 + \Gamma^w = 1 + \Gamma^y(1 - \chi n).$$

This proves the necessary restriction. $\square$

Proof of Proposition 4. Hold w^σ and χ fixed. Start with a female-wage increase and choose the marriage-benefit adjustment to keep aggregate fertility locally constant:

$$\frac{d\lambda}{dw^\varnothing} = -\frac{n_{w^\varnothing}}{n_\lambda} > 0.$$

The implicit function theorem applies because $n_\lambda > 0$. Household fertilities are independent of λ , so along this path

$$0 = \mathcal{M}(n^m)_{w^\varnothing} + (1 - \mathcal{M})(n^s)_{w^\varnothing} + (n^m - n^s) \frac{d\mathcal{M}}{dw^\varnothing}.$$

The first two terms are negative and $n^m > n^s$, implying $d\mathcal{M}/dw^\varnothing > 0$. The income gap falls because the female wage rises while female labor supply $l^\varnothing = 1 - \chi n$ remains constant.

Choose a slightly larger positive slope for λ . Aggregate fertility then rises strictly. Continuity preserves the strict marriage increase and income-gap decline for a sufficiently small change. Thus all three objectives improve.

The child-based measure improves as well. On the constant-fertility path it equals $1 - (1 - \mathcal{M})n^s/n$, which rises because marriage rises and single fertility falls. This strict increase persists under the sufficiently small additional marriage-benefit adjustment. $\square$

E Extended Model with Gender-Differentiated Childcare

Throughout this appendix, the father's raw time per child is χ^σ/η , where $\eta \in (0, 1]$ is paternal care efficiency. All formulas reduce to the original when $\eta = 1$.

E.1 Household Solutions and Proof of Lemma 4

Write the wife's monetary price of fertility as

$$P \equiv \tilde{w}^\varnothing \tilde{\chi}^\varnothing = w^\varnothing \chi^\varnothing + \alpha w^\sigma \frac{\chi^\sigma}{\eta}.$$

Throughout this appendix, household allocations are interior, $P > 0$, and both spouses have positive consumption and market labor supply. We retain z and R as defined in the main text.

Proof. The wife's budget is

$$c^{\varnothing,m} + Pn^m = \tilde{w}^\varnothing.$$

Her first-order condition gives

$$\frac{c^{\varnothing,m}}{n^m} = \left(\frac{(1-\beta)P}{\beta} \right)^\rho.$$

Consequently,

$$\begin{aligned} n^m &= \frac{\tilde{w}^\varnothing}{P + \left(\frac{(1-\beta)P}{\beta} \right)^\rho}, \\ c^{\varnothing,m} &= \frac{\tilde{w}^\varnothing}{1 + \left(\frac{\beta}{1-\beta} \right)^\rho P^{1-\rho}}. \end{aligned} \tag{E.1}$$

Using $P = \tilde{w}^\varnothing \tilde{\chi}^\varnothing$ in the fertility expression proves (14). Substituting the optimal allocation into CES utility gives

$$V^{\varnothing,m} = \lambda(1-\beta)^{1/r} \tilde{w}^\varnothing \left[1 + \left(\frac{\beta}{1-\beta} \right)^\rho P^{1-\rho} \right]^{1/(\rho-1)},$$

which proves (15). The singlehood formula follows by setting $\lambda = 1$, $\tilde{w}^\varnothing = w^\varnothing$, and $P = w^\varnothing \chi$.

At fixed $\tilde{w}^\varnothing$, fertility and indirect utility decrease with P . The consumption expression in (E.1) increases with P because $\rho > 1$. A lower fertility price therefore raises fertility and utility while lowering consumption. $\square$

E.2 Equilibrium Derivatives

Partial derivatives with respect to δ hold α fixed; primes denote total equilibrium derivatives. All expressions are evaluated at the initial equilibrium. At $\chi^{\sigma^\dagger} = 0$, policy derivatives are interpreted as right-hand derivatives.

Write the spouses' log marriage premiums as

$$f \equiv \log \frac{V^{\varnothing,m}}{V^{\varnothing,s}}, \quad h \equiv \log \frac{V^{\sigma^\dagger,m}}{V^{\sigma^\dagger,s}}.$$

Market clearing requires $f = h$. Recall that

$$z = \frac{P n^m}{\tilde{w}^\varnothing} \in (0, 1).$$

Differentiating the household fertility formula gives

$$\begin{aligned} n_\alpha^m &= n^m w^\sigma \left[\frac{1}{\tilde{w}^\varnothing} - \frac{\chi^\sigma / \eta}{P} (\rho - (\rho - 1)z) \right], \\ n_\delta^m &= \frac{n^m (w^\varnothing - \alpha w^\sigma / \eta)}{P} (\rho - (\rho - 1)z). \end{aligned} \quad (\text{E.2})$$

Here $P_\alpha = w^\sigma \chi^\sigma / \eta$ and $P_\delta = -(w^\varnothing - \alpha w^\sigma / \eta)$.

The wife's envelope condition yields

$$f_\alpha = \frac{w^\sigma (1 - \frac{\chi^\sigma}{\eta} n^m)}{\tilde{w}^\varnothing} > 0, \quad f_\delta = \frac{(w^\varnothing - \alpha w^\sigma / \eta) n^m}{\tilde{w}^\varnothing}. \quad (\text{E.3})$$

To see this, homogeneity and the wife's first-order condition imply

$$\frac{u_c(c^{\varnothing,m}, n^m)}{u(c^{\varnothing,m}, n^m)} = \frac{1}{\tilde{w}^\varnothing}.$$

For the husband, Euler's theorem and the definition of R imply

$$c^{\sigma,m} + \frac{u_n(c^{\sigma,m}, n^m)}{u_c(c^{\sigma,m}, n^m)} n^m = (1 - \alpha) w^\sigma + R n^m > 0.$$

His log-premium derivatives are therefore

$$\begin{aligned} h_\alpha &= \frac{-w^\sigma (1 - \frac{\chi^\sigma}{\eta} n^m) + R n_\alpha^m}{(1 - \alpha) w^\sigma + R n^m}, \\ h_\delta &= \frac{-(1 - \alpha) w^\sigma n^m / \eta + R n_\delta^m}{(1 - \alpha) w^\sigma + R n^m}. \end{aligned} \quad (\text{E.4})$$

Singlehood values remain fixed throughout this experiment.

Lemma (Local equilibrium responses). *At an interior equilibrium with $f_\alpha - h_\alpha \neq 0$, there is a differentiable local equilibrium branch. Writing $G \equiv f' = h'$ for the common log-premium*

response,

$$G = \frac{f_\alpha h_\delta - h_\alpha f_\delta}{f_\alpha - h_\alpha}, \quad (\text{E.5})$$

$$\mathcal{M}' = \theta \mathcal{M} (1 - \mathcal{M}) G, \quad (\text{E.6})$$

$$\alpha' = \frac{\tilde{w}^\varnothing G - (w^\varnothing - \alpha w^{\sigma^\nearrow} / \eta) n^m}{w^{\sigma^\nearrow} (1 - \frac{\chi^{\sigma^\nearrow}}{\eta} n^m)}, \quad (\text{E.7})$$

$$(n^m)' = n_\delta^m + n_\alpha^m \alpha'. \quad (\text{E.8})$$

The responses also satisfy

$$R(n^m)' = (w^{\sigma^\nearrow} + w^\varnothing + R n^m) G + (w^{\sigma^\nearrow} / \eta - w^\varnothing) n^m. \quad (\text{E.9})$$

Proof. Apply the implicit function theorem to $f - h = 0$. Differentiating market clearing gives

$$\alpha' = \frac{h_\delta - f_\delta}{f_\alpha - h_\alpha}.$$

Substitution into $G = f_\delta + f_\alpha \alpha'$ proves (E.5). Differentiating the participation rule proves (E.6). Solving the wife's envelope equation for α' gives (E.7), and the chain rule gives (E.8).

To obtain (E.9), write the spouses' total log-premium differentials as

$$\tilde{w}^\varnothing G = w^{\sigma^\nearrow} (1 - \frac{\chi^{\sigma^\nearrow}}{\eta} n^m) \alpha' + (w^\varnothing - \alpha w^{\sigma^\nearrow} / \eta) n^m$$

and

$$((1 - \alpha) w^{\sigma^\nearrow} + R n^m) G = -w^{\sigma^\nearrow} (1 - \frac{\chi^{\sigma^\nearrow}}{\eta} n^m) \alpha' - (1 - \alpha) w^{\sigma^\nearrow} n^m / \eta + R(n^m)'.$$

Adding these equations eliminates the transfer response. Since $\tilde{w}^\varnothing + (1 - \alpha) w^{\sigma^\nearrow} = w^\varnothing + w^{\sigma^\nearrow}$, and $(w^\varnothing - \alpha w^{\sigma^\nearrow} / \eta) - (1 - \alpha) w^{\sigma^\nearrow} / \eta = w^\varnothing - w^{\sigma^\nearrow} / \eta$, the result is (E.9). $\square$

Single fertility is fixed, so aggregate responses satisfy

$$n' = \mathcal{M}(n^m)' + \mathcal{M}'(n^m - n^s), \quad (\text{E.10})$$

and

$$(\mathcal{M} n^m)' = \mathcal{M}(n^m)' + \mathcal{M}' n^m. \quad (\text{E.11})$$

The child-based family-structure response is

$$\left(\frac{\mathcal{M}n^m}{n}\right)' = \frac{n^s [n^m \mathcal{M}' + \mathcal{M}(1 - \mathcal{M})(n^m)']}{n^2}.$$

Thus rising marriage and rising marital fertility imply an increase in the child-based counterpart.

E.3 The Male-Side Compensated Interpretation

Consider an auxiliary transfer adjustment that holds the wife's log marriage premium fixed. Denote this adjustment by α'_c . The wife's envelope condition gives

$$0 = f_\delta + f_\alpha \alpha'_c, \quad \alpha'_c = -\frac{(w^\varnothing - \alpha w^{\sigma^\varnothing}/\eta) n^m}{w^{\sigma^\varnothing} (1 - \frac{\chi^{\sigma^\varnothing}}{\eta} n^m)}.$$

Substituting into (E.2) gives the compensated fertility response:

$$\frac{(n^m)'_c}{n^m} = \frac{(w^\varnothing - \alpha w^{\sigma^\varnothing}/\eta) \rho(1 - z)}{P(1 - \frac{\chi^{\sigma^\varnothing}}{\eta} n^m)}.$$

The husband's compensated log-premium derivative is

$$\begin{aligned} h'_c &= h_\delta + h_\alpha \alpha'_c \\ &= \frac{n^m}{(1 - \frac{\chi^{\sigma^\varnothing}}{\eta} n^m)((1 - \alpha)w^{\sigma^\varnothing} + Rn^m)} \\ &\quad \times \left[\frac{R(w^\varnothing - \alpha w^{\sigma^\varnothing}/\eta) \rho(1 - z)}{P} - (1 - \frac{\chi^{\sigma^\varnothing}}{\eta} n^m)(w^{\sigma^\varnothing}/\eta - w^\varnothing) \right]. \end{aligned} \tag{E.12}$$

Its denominator is positive. Since $P = \tilde{w}^\varnothing \tilde{\chi}^\varnothing$, Condition (19) is exactly the requirement $h'_c > 0$.

This auxiliary path compensates the wife rather than clearing the marriage market. To connect it to equilibrium, use (E.5) to obtain

$$G = \frac{f_\alpha}{f_\alpha - h_\alpha} h'_c.$$

The proof below establishes $f_\alpha - h_\alpha > 0$ under Conditions (16) and (19). A positive compensated gain for the husband then raises the common equilibrium marriage premium.

E.4 Proof of Proposition 5

Proof. Condition (16) gives $w^{\varnothing} - \alpha w^{\sigma} / \eta > 0$. Together with Condition (19), $w^{\sigma} > w^{\varnothing}$, and interiority, this implies $R > 0$.

Using (E.2)–(E.4),

$$\begin{aligned} f_{\alpha} - h_{\alpha} &= \frac{w^{\sigma}}{\tilde{w}^{\varnothing}((1 - \alpha)w^{\sigma} + Rn^m)} \\ &\times \left[(w^{\sigma} + w^{\varnothing})\left(1 - \frac{\chi^{\sigma}}{\eta} n^m\right) + \frac{R \frac{\chi^{\sigma}}{\eta} n^m \tilde{w}^{\varnothing} \rho(1 - z)}{P} \right] > 0. \end{aligned} \quad (\text{E.13})$$

Thus the equilibrium is locally regular.

Condition (19) implies $h'_c > 0$. Since $f_{\alpha} > 0$ and $f_{\alpha} - h_{\alpha} > 0$, the compensated comparison gives

$$G = \frac{f_{\alpha}}{f_{\alpha} - h_{\alpha}} h'_c > 0.$$

Equation (E.6) therefore gives $\mathcal{M}' > 0$. Singlehood values are fixed, so both spouses' marital values and marriage-to-singlehood utility ratios rise.

Marital fertility rises directly from (E.9):

$$R(n^m)' = (w^{\sigma} + w^{\varnothing} + Rn^m)G + (w^{\sigma} / \eta - w^{\varnothing})n^m > 0.$$

Because $R > 0$, this implies $(n^m)' > 0$.

Under fixed total care,

$$P = w^{\varnothing}\chi - (w^{\varnothing} - \alpha w^{\sigma} / \eta)\chi^{\sigma} \leq w^{\varnothing}\chi, \quad \tilde{w}^{\varnothing} > w^{\varnothing}.$$

Fertility increases with resources and decreases with its monetary price. Comparing the married and single household problems therefore gives $n^m > n^s$. Equation (E.10) now implies $n' > 0$.

Differentiating aggregate labor supplies gives

$$(l^{\sigma})' = - \left[\frac{1}{\eta} \mathcal{M}n^m + \frac{\chi^{\sigma}}{\eta} (\mathcal{M}n^m)' \right] < 0$$

and

$$(l^{\varnothing})' = \mathcal{M}n^m + \chi^{\sigma} (\mathcal{M}n^m)' - \chi n'.$$

At fixed wages,

$$\begin{aligned}\frac{d \log \Gamma^y}{d\delta} &= \frac{(l^{\sigma'})'}{l^{\sigma'}} - \frac{(l^{\varphi})'}{l^{\varphi}} \\ &= \frac{\chi n'}{l^{\varphi}} - \left[\mathcal{M}n^m + \chi^{\sigma'} (\mathcal{M}n^m)' \right] \left(\frac{1}{\eta l^{\sigma'}} + \frac{1}{l^{\varphi}} \right).\end{aligned}\tag{E.14}$$

Condition (18) is exactly the requirement that this expression be negative. It does not require female labor supply to rise.

The strict derivative signs and interiority imply the same directional changes for sufficiently small feasible reforms. The child-based family-structure counterpart also rises, because marriage and marital fertility both increase. $\square$

E.5 Redistribution Starting from the Baseline

At $\chi^{\sigma'} = 0$ and $\chi^{\varphi} = \chi$,

$$P = w^{\varphi} \chi, \quad R = \frac{\beta}{1 - \beta} \left(\frac{(1 - \alpha)w^{\sigma'}}{n^m} \right)^{1/\rho}, \quad z = \frac{w^{\varphi} \chi n^m}{w^{\varphi} + \alpha w^{\sigma'}}.$$

Substitution into the equilibrium derivatives gives

$$G = \frac{n^m}{w^{\sigma'} + w^{\varphi}} \left[\frac{R(w^{\varphi} - \alpha w^{\sigma'} / \eta) \rho(1 - z)}{w^{\varphi} \chi} - (w^{\sigma'} / \eta - w^{\varphi}) \right], \tag{E.15}$$

and

$$\frac{(n^m)'}{n^m} = G + \frac{(w^{\varphi} - \alpha w^{\sigma'} / \eta) \rho(1 - z)}{w^{\varphi} \chi}. \tag{E.16}$$

The male-side condition becomes

$$R(w^{\varphi} - \alpha w^{\sigma'} / \eta) \rho(1 - z) > w^{\varphi} \chi (w^{\sigma'} / \eta - w^{\varphi}).$$

Since $l^{\sigma'} = 1$ initially, the income gap narrows if and only if

$$\chi n' < \frac{\mathcal{M}n^m}{\eta} (\eta + l^{\varphi}).$$

Marriage and aggregate fertility derivatives follow from (E.6) and (E.10).

E.6 Proof of Proposition 6

Proof. The maintained regularity condition is $f_\alpha - h_\alpha > 0$.

Case (A). If $w^\varphi - \alpha w^\sigma / \eta \leq 0$, (E.2) and (E.3) imply $n_\delta^m \leq 0$ and $f_\delta \leq 0$.

Starting from the baseline, $R > 0$ and $w^\sigma / \eta > w^\varphi$. Thus (E.15) gives $G < 0$, including when $w^\varphi = \alpha w^\sigma / \eta$. Consequently, $\mathcal{M}' < 0$, and (E.16) gives $(n^m)' < 0$. Baseline marital fertility exceeds single fertility, so (E.10) implies $n' < 0$.

At the baseline, (E.14) reduces to

$$\frac{d \log \Gamma^y}{d\delta} = \frac{\chi n'}{l^\varphi} - \frac{\mathcal{M} n^m}{\eta} \left(1 + \frac{\eta}{l^\varphi} \right) < 0.$$

The child-based family-structure counterpart falls because marriage and marital fertility both fall.

With positive paternal care initially, the fixed-transfer signs remain valid, but additional restrictions are needed to determine the full equilibrium responses.

Case (B). Here $f_\delta > 0$ and $h_\delta \leq 0$. Hence

$$\alpha' = \frac{h_\delta - f_\delta}{f_\alpha - h_\alpha} < 0.$$

The marriage response is determined by (E.5)–(E.6), and marital fertility responds according to (E.8). These formulas include both the direct policy effect and the equilibrium transfer adjustment.

Case (C). Conditions (16) and (19) imply rising marriage and fertility. Failure of Condition (18) means

$$\chi n' \geq \left[\mathcal{M} n^m + \chi^\sigma (\mathcal{M} n^m)' \right] \left(1 + \frac{l^\varphi}{\eta l^\sigma} \right).$$

Equation (E.14) then gives $d \log \Gamma^y / d\delta \geq 0$. □

E.7 Different Total Childcare Requirements by Marital Status

The household and equilibrium derivative formulas remain valid if $\chi^\varphi + \chi^\sigma \neq \chi$, provided both total requirements remain fixed during redistribution and household choices remain interior.

The aggregate fertility conclusion additionally requires $n^m \geq n^s$. Together with increasing marriage and strictly increasing marital fertility, this is sufficient for $n' > 0$ by (E.10).

Labor supplies are

$$l^{\sigma} = 1 - \frac{\chi^{\sigma}}{\eta} \mathcal{M}n^m, \quad l^{\varphi} = 1 - \chi^{\varphi} \mathcal{M}n^m - \chi(1 - \mathcal{M})n^s.$$

Their derivatives are

$$(l^{\sigma})' = -\frac{1}{\eta} \mathcal{M}n^m - \frac{\chi^{\sigma}}{\eta} (\mathcal{M}n^m)',$$

and

$$(l^{\varphi})' = \mathcal{M}n^m - \chi^{\varphi} (\mathcal{M}n^m)' + \chi n^s \mathcal{M}'.$$

The exact condition for a declining income gap is

$$\frac{\mathcal{M}n^m - \chi^{\varphi} (\mathcal{M}n^m)' + \chi n^s \mathcal{M}'}{l^{\varphi}} + \frac{\frac{1}{\eta} \mathcal{M}n^m + \frac{\chi^{\sigma}}{\eta} (\mathcal{M}n^m)'}{l^{\sigma}} > 0.$$

The child-based family-structure counterpart still rises whenever marriage and marital fertility rise, because single fertility remains fixed.

F Dynamic Extension: Cross-Generational Feedback

This appendix extends the static model of Section 3 by allowing current family structure to shape the next generation's gender wage gap.

The static model takes the gender human-capital gap as an input. Yet family structure may itself shape that gap in the next generation. A substantial literature links children's outcomes to their family environment and finds that boys are often more sensitive than girls to disadvantage and the absence of a biological father (Sommers, 2001; Bertrand and Pan, 2013; Chetty et al., 2016; Autor et al., 2019; Wasserman, 2020; Reeves, 2022; Frimmel et al., 2024). Autor et al. (2019) examine gender differences associated with family disadvantage, and Autor et al. (2023) study family structure and high-school performance.

These findings suggest a feedback worth exploring. If lower marriage weakens future male human capital relative to female human capital, it changes the earning opportunities that support the next generation's marital contracts. Today's family choices can then affect tomorrow's marriage market.

I introduce a reduced-form law of motion:

$$\Gamma_{t+1}^w = \mathcal{H}(\mathcal{M}_t), \quad \mathcal{H}'(\mathcal{M}_t) > 0. \quad (\text{F.1})$$

Marriage serves as a summary predictor of the family environment. Equation (F.1) captures boys'

greater sensitivity to that environment, allowing the underlying mechanism to operate through parental inputs, role models, or peer and neighborhood effects. The relationship is imposed as a modeling assumption; establishing its causal magnitude and its connection to children's living arrangements would require separate empirical analysis.

Figure A.5 brings the channels together. The human-capital link runs across generations; time subscripts are omitted from the diagram to keep the mechanism visible.

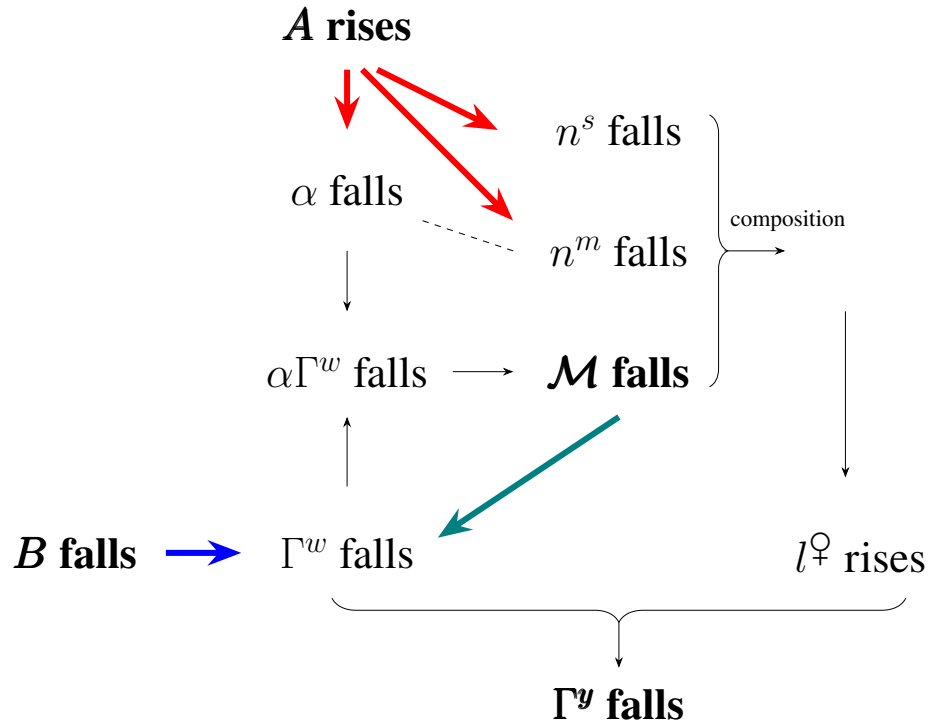


Figure A.5. The Dynamic Mechanism

Notes: Red arrows represent the effects of rising gender-neutral productivity A_t . The blue arrow represents the effect of falling gender-biased productivity B_t . The teal arrow represents the dynamic human-capital effect through $\mathcal{M}_t \rightarrow \Gamma_{t+1}^w$ in Equation (F.1). Time subscripts are suppressed. The diagram summarizes directions of propagation; their magnitudes and the stability of the system depend on the response elasticities.

The transition unfolds through two exogenous drivers and one endogenous feedback:

- (1) **Gender-neutral productivity growth** raises the opportunity cost of children. Fertility falls within both marital states, and the equilibrium transfer and income-sharing multiplier decline. Marriage becomes less attractive, so the shift toward singlehood reinforces lower aggregate fertility. Women devote less time to childcare, and the income gap narrows.
- (2) **Declining gender-biased productivity** improves women's wages at a given male wage. Children become more costly in forgone female earnings, while the transfer adds less relative to women's own resources. Fertility and marriage fall, and the income gap narrows through both wages and market work.

- (3) **Dynamic propagation** carries the marriage response into the next generation. Lower current marriage reduces future male relative human capital through (F.1). This weakens the income-sharing basis of future marriages and puts additional downward pressure on fertility. By Lemma 3, the resulting decline in Γ^w also changes the composition of the trinity shortfall: as the wage gap narrows, the fertility–gender equality tension loosens while the fertility–marriage tension tightens, potentially shifting the binding constraint across generations.

Whether the feedback in channel (3) amplifies or merely prolongs the demographic response depends on quantitative features that the sign restriction in (F.1) alone does not pin down. A steep $\mathcal{H}(\cdot)$ combined with highly responsive marriage participation could generate persistent amplification, while a flat $\mathcal{H}(\cdot)$ or a weak marriage-participation elasticity would produce a feedback that decays rapidly. Characterizing the boundary between these regimes requires specifying the functional form of $\mathcal{H}$ and estimating its slope, which in turn demands data linking family structure to children’s subsequent human capital accumulation, a task beyond the scope of the present paper.

Two qualitative implications are nonetheless worth noting. First, both gender-neutral and gender-biased technological change can simultaneously depress fertility, erode marriage, and narrow gender income gaps, because all three outcomes are tied together by the same equilibrium system. This complements earlier emphases on factor-biased technical change in Greenwood et al. (2005b) and Ngai and Petrongolo (2017). Second, policies that change marriage today may also change the distribution of earning opportunities tomorrow. Their long-run importance depends on the strength of the human-capital link together with the immediate demographic response, and quantifying that importance remains an open question.

G Robustness of the Quantitative Results to the Childcare Time Cost

The baseline calibration sets the childcare time cost at $\chi = 0.075$, following [De La Croix and Doepke \(2003\)](#). Because this parameter is not estimated from Mexican micro-data, its value warrants a sensitivity check. This appendix re-does the entire quantitative exercise of Section 5 for $\chi \in \{0.05, 0.10\}$, bracketing the baseline. For each value of χ , I re-recover wages from the accounting identities, re-estimate the four preference and marriage parameters $(\beta, \rho, \lambda, \theta)$ to match fertility and dual parenthood, and re-solve the equilibrium in every year. All other data inputs—GDP per capita, female income shares, and the ENUT-based childcare division—remain unchanged.

Table A.7 reports the re-estimated parameters for each specification alongside the baseline.

χ	β	ρ	λ	θ
0.050	0.050	2.50	0.979	15
0.075 (baseline)	0.077	2.50	0.954	12
0.100	0.100	2.80	0.929	12

Table A.7. Re-Estimated Parameters Under Alternative χ

Notes: For each value of χ , the four parameters $(\beta, \rho, \lambda, \theta)$ are re-estimated to match Mexico’s fertility and dual-parenthood series over 1990–2025. The middle row reproduces Table 6.

The re-estimated parameters adjust in intuitive directions. A lower χ mechanically raises fertility at given preferences, so the estimation responds by lowering the fertility weight β (from 0.077 to 0.050) to match the same observed fertility levels. The marriage benefit λ rises (from 0.954 to 0.979), because the smaller time cost reduces the marital fertility premium and hence the income-sharing attraction of marriage; a higher λ compensates by raising the common benefit. Participation responsiveness θ increases from 12 to 15, producing sharper marriage adjustments to offset the attenuated fertility channel. When $\chi = 0.10$, the opposite pattern obtains: β rises to 0.100 and λ falls to 0.929, while ρ increases from 2.50 to 2.80 to accommodate the stronger substitution effect implied by the larger time cost. All three specifications remain firmly in the substitution-dominant region ($\rho > 1$).

Figure A.6 reports the results across six panels. Three findings emerge.

Model fit is preserved. Panels (a) and (b) overlay the data with model predictions under each χ . All three specifications track the observed fertility decline and the level and trend of dual parenthood. The fit differences across specifications are modest relative to the overall magnitude of the transitions.

The gender income gap and welfare trajectories are nearly invariant. Panel (c) shows that the model-implied gender income gap is virtually identical under the three specifications. This insensitivity reflects the fact that the income gap is driven primarily by the exogenous wage-gap input B_t , which is common across specifications. The direct effect of χ on female labor supply is a second-order consideration once wages have been re-recovered to match the same female income shares. Panel (d) shows that the consumption-equivalent welfare gap for married women shifts in level—higher χ implies a larger welfare gap, because the time cost of children reduces the wife’s effective resources more severely—but the declining trend driven by wage convergence is preserved across all three cases.

The scope for redistribution widens with lower χ , and the Shapley decomposition is qualitatively stable. Panel (e) plots the minimum paternal efficiency threshold $\bar{\eta}$ from equation (20). The threshold is sensitive to χ : at $\chi = 0.05$, $\bar{\eta}$ falls to approximately 0.65 by the mid-2010s, compared with 0.77 at the baseline and roughly 0.85 at $\chi = 0.10$. The intuition is that a lower time cost per child reduces the father’s foregone earnings when he provides care, relaxing Condition (19) and widening the set of paternal efficiencies under which redistribution improves all three outcomes. Importantly, all three specifications exhibit the same qualitative pattern: $\bar{\eta}$ declines over the sample as the wage gap narrows, confirming that the expanding scope for redistribution documented in Section 5.4 is not an artifact of the baseline χ .

Panel (f) reports the Shapley-value decomposition of fertility for each specification. The dominant role of the declining wage gap (B , maroon) is preserved across all three values of χ . The offsetting contribution of the childcare-division channel (χ -force, green) grows in magnitude with the level of χ , as expected: when children are more time-intensive, changes in the gender division of that time exert a larger effect on fertility. Gender-neutral productivity (A , navy) plays a small supporting role throughout. These patterns confirm that the qualitative conclusions of the Shapley decomposition in Section 5.5—wage convergence as the principal driver, childcare redistribution as a partial buffer—are robust to the assumed level of childcare costs.

In summary, the quantitative conclusions of Section 5 are robust to plausible variation in the childcare time cost. The model continues to fit the targeted moments, the Shapley decomposition maintains its qualitative structure, and the declining paternal-efficiency threshold—the paper’s central quantitative finding on the scope for childcare redistribution—is preserved across all three specifications.

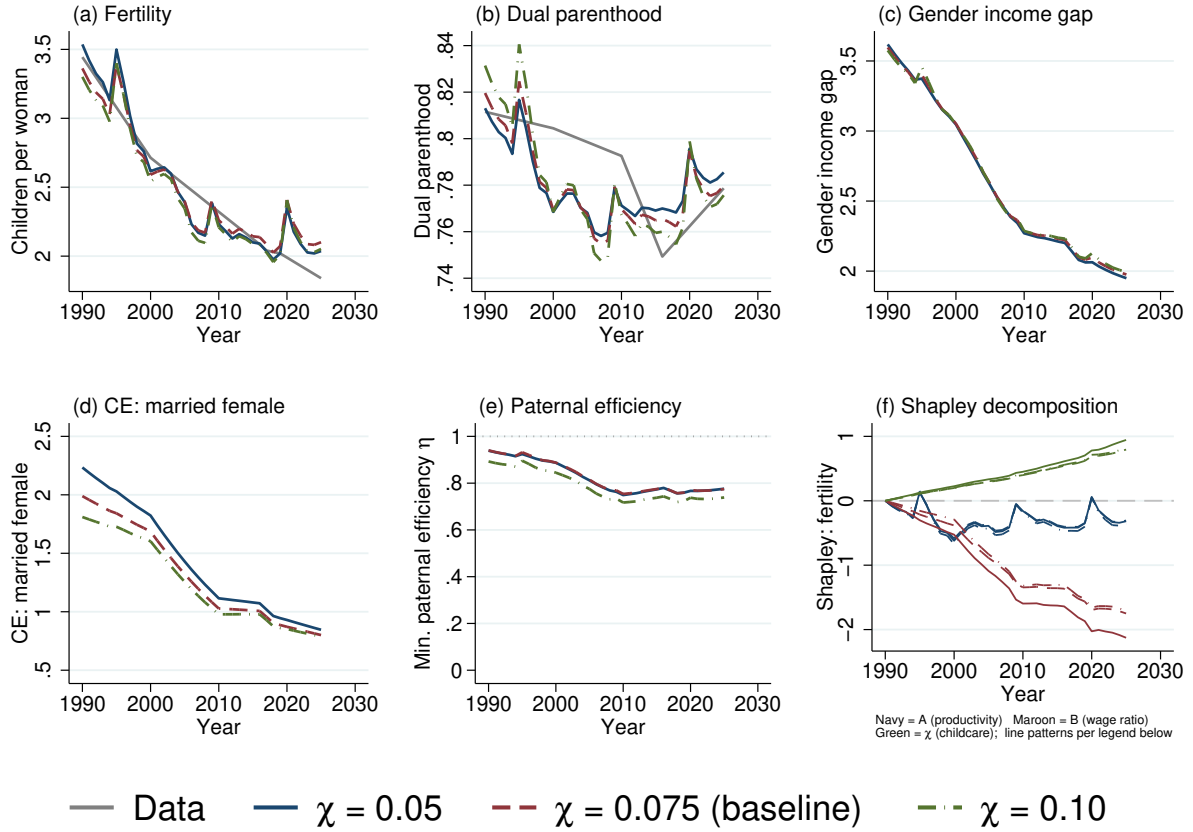


Figure A.6. Robustness of the Calibration to Alternative Values of χ

Notes: Each panel compares the calibrated model under $\chi = 0.05$ (solid navy), $\chi = 0.075$ (dashed maroon, baseline), and $\chi = 0.10$ (dash-dot green), using the parameters in Table A.7. Panels (a) and (b) also plot the data (solid gray). For each χ , wages are re-recovered and the four parameters ($\beta, \rho, \lambda, \theta$) are re-estimated to match fertility and dual parenthood. Panel (c): gender income gap Γ^y . Panel (d): consumption equivalent for married females benchmarked to the married male. Panel (e): minimum paternal efficiency threshold $\bar{\eta}$ from (20); the dotted line at $\eta = 1$ marks equal efficiency. Panel (f): Shapley-value decomposition of fertility; colours encode the force (A = navy, B = maroon, χ = green) and line patterns encode the χ -scenario as in the shared legend.

H Why the Wage Gap Determines the Dominant Tension

Lemma 3 identifies the gender wage gap Γ^w as the key state variable that determines whether the F–G or F–M tension dominates the trinity shortfall. OECD and non-OECD countries differ not only in Γ^w , however, but also in wage levels, fertility preferences, and the gender division of childcare. This appendix asks whether any of these alternatives could independently generate the empirical pattern in Table 4, and argues that Γ^w occupies a structurally privileged position within the model. The structural comparisons below hold at any interior equilibrium; the quantitative magnitudes cited from the Shapley decomposition are computed under parameters calibrated to Mexico (Table 6) and would need to be re-evaluated for other economies.

Gender-neutral productivity (A). Consider two economies sharing $(\beta, \rho, \chi, \lambda, \theta)$ and Γ^w but differing in gender-neutral productivity $A = w^\varnothing$ (with $w^\varnothing = A/\Gamma^w$). The F–G ceiling defined in Lemma 3(i),

$$\bar{n}(\Gamma^w) = \frac{1}{\chi} \left(1 - \frac{\Gamma^w}{\bar{\Gamma}} \right),$$

depends on Γ^w , χ , and the equality threshold $\bar{\Gamma}$, but not on A . This is immediate: the accounting identity $\Gamma^y = \Gamma^w/(1 - \chi n)$ is independent of wage levels. The fertility ceiling compatible with a given degree of gender equality is therefore the same at every productivity level, provided the wage gap is the same.

Productivity does affect the F–M coupling. A higher A raises C (since $\rho > 1$ implies that $(w^\varnothing)^{\rho-1}$ increases with A), which raises K and hence $g = K^{1/r}$. Since $\partial(q - 1)/\partial g < 0$ (Appendix D.5), the marital fertility premium $q - 1$ falls. Higher productivity thus weakens the coupling between marriage and fertility, but it operates through a single channel: $A \rightarrow g \rightarrow q$. By contrast, Γ^w enters the coupling through two reinforcing channels—directly through the Γ^w terms in $q - 1 = (1 - g)\Gamma^w/(1 + g\Gamma^w)$ and indirectly through g —as established in Lemma 3(ii).

The asymmetry is therefore clean: A can modulate the F–M coupling through one channel but is irrelevant for the F–G amplification, whereas Γ^w enters both mechanisms simultaneously. The Shapley-value decomposition of Mexico’s income-gap convergence (Table 8) provides quantitative corroboration: gender-neutral productivity A accounts for 3.3 percent of the total change in Γ^y , compared with 94.5 percent for the relative wage gap B . The income-gap result reflects that a proportional rise in productivity raises both male and female wages similarly, leaving relative earnings nearly unchanged.

Fertility preferences (β). The F–G ceiling $\bar{n}(\Gamma^w)$ is derived from the accounting identity $\Gamma^y = \Gamma^w/(1 - \chi n)$ and is therefore independent of preferences. Different values of β change realized fertility and consequently determine whether an economy’s equilibrium fertility exceeds the ceiling,

but they cannot move the ceiling itself. Attributing the differential bottleneck to β would amount to restating the observation—countries with fertility above the ceiling face the F–G tension—rather than explaining it. Within the model, β affects the level of fertility but does not alter the structural relationship between Γ^w and the two mechanisms that Lemma 3 identifies.

Gender division of childcare. In the extended model of Section 4, both parents provide childcare and the income gap becomes $\Gamma^y = \Gamma^w \cdot l^{\sigma} / l^{\varphi}$. When fathers share some of the childcare ($\chi^{\sigma} > 0$), the F–G relationship changes in two specific ways relative to the baseline.

First, the ceiling on fertility compatible with a given degree of gender equality relaxes. With paternal care, male labor supply falls below unity ($l^{\sigma} = 1 - \frac{\chi^{\sigma}}{\eta} \mathcal{M} n^m < 1$), while female labor supply rises relative to the baseline ($l^{\varphi} = 1 - \chi n + \chi^{\sigma} \mathcal{M} n^m > 1 - \chi n$). The ratio l^{σ} / l^{φ} is therefore smaller than $1 / (1 - \chi n)$, and the maximum fertility consistent with $\Gamma^y \leq \bar{\Gamma}$ increases. The magnitude of this shift, however, is modest at observed childcare divisions. Under the calibrated Mexican parameters at 2025— $\chi^{\sigma} \approx 0.020$, $\mathcal{M} \approx 0.77$, $n^m \approx 1.8$, and $\eta = 1$ —the male labor-supply reduction is approximately $\chi^{\sigma} \mathcal{M} n^m \approx 0.028$, a roughly 3 percent departure from full market time. This is small relative to the female childcare burden $\chi n \approx 0.135$ that drives the main amplification, leaving the F–G ceiling primarily determined by Γ^w .

Second, the F–M coupling is also affected. With shared care, the marital fertility premium $q = n^m / n^s$ depends on the wife’s effective childcare cost $\tilde{\chi}^{\varphi}$ (equation 13), which itself depends on the childcare division. At a fixed transfer, shifting care toward the husband lowers $\tilde{\chi}^{\varphi}$ and raises n^m , increasing the premium. The equilibrium transfer adjusts in response, however, and the net effect on q depends on the full set of equilibrium derivatives (Appendix E.2). Unlike the two-channel characterization available for Γ^w in Lemma 3(ii), the childcare channel’s effect on the F–M coupling does not admit a simple directional summary without specifying the economic environment.

The Shapley decomposition confirms the quantitative asymmetry. The childcare-division force accounts for -63.1 percent of the fertility change and -167.4 percent of the dual-parenthood change, but only 2.2 percent of the income-gap change (Table 8). The childcare channel thus operates primarily through the household margin—lowering the effective price of children within marriage, which raises n^m and supports $\mathcal{M}$ —while the income-gap margin remains dominated by the direct wage effect of Γ^w .

Summary. Table A.8 collects the comparisons.

The gender wage gap is the only variable that enters both the F–G amplification and the F–M coupling through direct, analytically characterized channels. Wage levels affect only the coupling and through a single indirect channel. Fertility preferences cannot move the ceiling. The childcare

Candidate	Affects F–G ceiling?	Affects F–M coupling?	Shapley share (Γ^y)
Γ^w (wage gap)	Yes, directly	Yes, two channels	94.5%
A (wage level)	No	Yes, one channel	3.3%
β (fertility preference)	No	Indirectly (through n)	—
χ^σ (childcare division)	Yes, modestly	Yes, direction ambiguous	2.2%

Table A.8. Alternative Candidates for Explaining the Differential Bottleneck

Notes: Each row reports whether the candidate variable affects the F–G ceiling (Lemma 3(i)), whether it affects the F–M coupling (Lemma 3(ii)), and its Shapley share of the income-gap change in the Mexico calibration (Table 8). The F–G ceiling for A and β refers to the baseline accounting identity $\Gamma^y = \Gamma^w / (1 - \chi n)$; for the childcare division it refers to the extended-model identity $\Gamma^y = \Gamma^w \cdot l^\sigma / l^\varphi$. The first two columns follow from the model’s analytical structure and hold at any interior equilibrium. The Shapley shares are computed under parameters calibrated to Mexico (Table 6); their magnitudes may differ for other economies. β does not vary in the Shapley decomposition and is therefore marked “—”.

division can relax the ceiling and influence the coupling, but its quantitative effects on income convergence are small—accounting for only 2.2 percent of the income-gap change in the Mexico calibration—and its effect on the coupling lacks a simple directional characterization. The structural argument that Γ^w simultaneously governs both mechanisms follows from the model’s analytical properties and holds at any interior equilibrium; the Shapley magnitudes provide quantitative corroboration from one calibrated economy and would need to be re-evaluated for others.